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29 April 1988
SUPERSEDING
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12 November 1981

MILITARY SPECIFICATION

MODEM, CABLE DRIVER, MD-1023(P)/G

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of Modem, Cable Driver, MD-1023(P)/G. This unit is also known as Low Speed Cable Driver Modem (LSCDM) and is one element of the Digital Group Multiplex (DGM) family of cable system equipments developed by the Joint Tactical Command, Control and Communications Agency (JTC^{3}A, formerly TRI-TAC). See 6.1 for intended use of the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specificatons, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Method of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-C-55583	Cable Assembly, Special purpose, Electrical CX-11230( )/G and Cable Assembly Adapter CX-10734( )/G
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Suvivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-B-875934	Modem, Cable Driver, MD-1023(P)/G
SM-D-875948	Circuit Card Assembly (CCA) BITE 1
SM-D-875979	CCA, Orderwire Transmitter
SM-D-875980	CCA,Orderwire Receiver
SM-D-875985	CCA, Traffic Receiver A Low Speed
SM-D-875986	CCA, Traffic Transmitter Low Speed
SM-D-876265	CCA, Traffic Receiver B Low Speed
SM-D-884567	CCA, Multi VDC
SM-D-884575	CCA, DC Input
SM-D-884579	CCA, AC Input
SM-D-884598	CCA, 400/800 VDC
SM-D-884678	CCA, AC Only

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-SM-B-875934, and the requirements specified herein.

3.2.2 Printed wiring boards. The printed wiring boards shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirement 30 and 64.

3.3 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and drawing DL-SM-B-875934.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.16) A unique serial number is required on each nomenclatured item and CCA. They shall be consecutively assigned.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each MD-1023(P)/G assembly shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.8 after burn-in.

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3.7 Electrical. (See 4.7).

3.7.1 Balanced equipment-side signal characteristics.

3.7.1.1 Line-to-line voltage. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1 and between -0.1 and -4.0 volts for a digital 0, measured into a 100 ohm dummy load. For the 3 kb/s data and timing outputs, the minimum voltages measured into a 100 ohm dummy load shall be +0.5 volts for a digital 1 and -0.5 volts for a digital 0.

3.7.1.2 Signal balance. (See 4.7.1.2) The magnitude of the difference between the magnitudes of the digital 1 level and digital 0 level shall be less than 10% of the larger of the two.

3.7.1.3 Line-to-ground voltage. (See 4.7.1.3) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.4 Clock/data phasing. (See 4.7.1.4) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be within ±29 ns for group rate signals, or ±1500 ns for 2 and 16 kb/s signals. Positive values mean timing leads data. Times shall be measured at the zero voltage point of the signals.

3.7.1.5 Offset. (See 4.7.1.5) The magnitude of the offset voltage measured between the center tap of a 100 ohm dummy load and ground shall not be greater than 3 volts.

3.7.1.6 Slew rate. (See 4.7.1.6) When 100 feet of RG-108 A/U cable terminated in a 78 ohm dummy load is connected to the driver output, the transition between ± 40 mV (either direction) shall take no more than 10 ns for group rate signals and 100 ns for the 2 and 16 kb/s signals.

3.7.1.7 Receiver dynamic range. (See 4.7.1.7) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV to +4 volts for a digital 1 or -50 mV to -4 volts for a digital 0 is applied to the receiver input terminals.

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3.7.1.8 Input load impedance. (See 4.7.1.8) The input load impedance shall be 100 ohms ±10%.

3.7.1.9 Clock asymmetry. (See 4.7.1.9) The positive half of the clock waveform shall be equal to half the period ±15 ns for group rate timing signals and ±1500 ns for 2 and 16 kb/s timing signals.

3.7.1.10 Unloaded output. (See 4.7.1.10) For the 2 kb/s data and timing outputs, the maximum voltages measured into an open circuit shall be +3.0 volts for a digital 1 and -3.0 volts for a digital 0.

3.7.2 Cable-side group signal characteristics. (See 4.7.2)

3.7.2.1 Output voltage levels. (See 4.7.2.1) The transmitter output voltage level shall be 2.0 V peak-to-peak ±20%.

3.7.2.2 Rise and fall times. (See 4.7.2.2) The rise and fall times shall be from 35 to 150 ns.

3.7.2.3 Receiver input impedance. (See 4.7.2.3) The receiver imput impedance shall be between 20 and 78 ohms over a frequency range from 500 to 2304 kHz.

3.7.2.4 Overshoot, undershoot and droop. (See 4.7.2.4) The overshoot and undershoot shall not exceed 15% of the peak-to-peak signal amplitude. The droop measured on a digital 0 shall not exceed 10% of the peak-to-peak signal amplitude.

3.7.3 Equipment-side maintenance orderwire (OW) signal characteristics. (See 4.7.3)

3.7.3.1 Input impedance. (See 4.7.3.1) The input impedance shall be from 540 to 660 ohms between 300 and 1700 Hz.

3.7.3.2 Signal-to-distortion ratio. (See 4.7.3.2) The signal-to-distortion ratio shall be 30 dB minimum at 1 kHz with a -4 dBm input signal.

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3.7.3.3 Output impedance. (See 4.7.3.3) The output impedance shall be 540 to 660 ohms between 300 and 1700 Hz.

3.7.3.4 Frequency response. (See 4.7.3.4) The frequency response with or without pulse restorers shall be as follows:

Frequency, Hz	dB
300	-3 to +6
1000	Reference (0dB)
1700	-3 to +3

3.7.4 Cable-side maintenance orderwire signal characteristics. (See 4.7.4)

3.7.4.1 Output power level. (See 4.7.4.1) The output power level shall be from 21.8 to 26 dBm with a -4 dBm analog voice orderwire (AVOW) input signal.

3.7.4.2 Input impedance. (See 4.7.4.2) The input impedance shall be from 260 to 900 ohms including the loading coil over the frequency range from 300 to 1700 Hz.

3.7.4.3 Output impedance. (See 4.7.4.3) The output impedance shall be from 260 to 900 ohms including the loading coil over the frequency range from 300 to 1700 Hz.

3.7.5 Frame acquisition time. (See 4.7.5) The modem shall acquire frame in 2.5 ms, or less, in 90% of the trials.

3.7.6 Prime power interface. (See 4.7.6)

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3.7.6.1 Power variation. (See 4.7.6.1) The dc primary power shall have a steady state voltage of 28 (+4, -6) volts. The ac primary power source shall be 115 V ±10% at frequencies of 50 Hz, 60 Hz, and 400 Hz ±5% single phase. The power input to the Cable Driver Modem MD-1023(P)/G shall not exceed 150 watts. The ac primary power shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Mode I, II and III generator sets.

3.7.6.2 Power feed. (See 4.7.6.2) Power for cable system operation shall be supplied by the power supply in the unit. It shall operate by a separate front panel ON/OFF switch and shall have the capacity to provide a constant current of 40 mA dc (nominal) to provide power for up to 39 restorer halves. The cable interface circuits shall withstand without damage to the equipment an injected current of 90 milliamperes on either center conductor of the CX-11230 cable. The unit shall be capable of operating with Power Feed switch in the OFF position.

3.7.6.3 Fault locate. (See 4.7.6.3) The equipment shall provide the capability to locate a faulty pulse form restorer (TD-1218/G) in a cable system with an accuracy of ±1 restorer using CX-11230 cable in accordance with MIL-C-55583.

3.7.6.4 Safety. (See 4.7.6.4) The equipment shall discharge the cable voltage to a value of 30 V or less in less than 2.0 seconds after the cable current is interrupted.

3.7.7 BITE and alarms. (See 4.7.7) The BITE shall monitor signals in and out of the unit and internal circuits, and power supply voltages. The BITE shall operate without interrupting traffic or interfering with normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for loss of frame sync, channel failure, common power supply failure, and interunit signal malfunctions. Alarm circuits shall be operational within 1 second of application of power to the unit. Alarm conditions are removed automatically when proper operation is restored.

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3.7.7.1 Alarms. (See 4.7.7.1) Failures detected by the BITE circuitry shall cause relay contact closures accessible externally at a connector on the rear panel. The following outputs shall be available.

a. S Alarm. The S Alarm shall indicate any failure detected including loss of input power. This alarm also provides a visible indication on the front panel except for loss of input power.

b. SA Alarm. The SA Alarm shall indicate that the malfunction is within the unit. Loss of input cable signal inhibits the SA Alarm. If the power switch is set to OFF, there shall be no SA Alarm.

c. LS Alarm. The LS Alarm shall indicate loss of signal coming in on CX-11230 cable. If the unit loses power, the LS Alarm assumes the normal state. This alarm inhibits the SA Alarm.

d. Lamp test. A front panel switch shall be provided in the unit to check the operation of the alarms without interrupting the equipment operation.

3.7.8 Operational traffic rate test. (See 4.7.8)

3.7.8.1 Group error check with ring tone. (See 4.7.8.1) With the ring tone present, there shall be no errors for a measurement period of 10 seconds at all data rates.

3.7.8.2 Digital voice orderwire error check with ring tone. (See 4.7.8.2) With the ring tone present, there shall be no errors for a measurement period of 30 seconds.

3.7.8.3 Data orderwire error check with ring tone. (See 4.7.8.3) With the ring tone present, there shall be no errors for a measurement period of 30 seconds.

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3.7.8.4 Alarm monitor operation. (See 4.7.8.4)

a. Normal alarm . No alarms shall be generated when the equipment is operated normally, that is, there shall be open circuits on the S, SA, and LS alarm outputs (a short indicates alarm).

b. Loss of input (LOI) alarm . Removal of the group signal to the equipment shall cause an S alarm and illuminate the LOI lamp.

c. Loss of signal alarm . Removal of the cable line signal to the equipment shall cause an LS alarm.

d. Alarm test operation. Activation of the lamp test switch shall cause an S, SA and LS alarm and illuminate the LOI lamp.

3.7.8.5 Maintenance orderwire output level. (See 4.7.8.5) The maintenance orderwire output level with -4 dBm input shall be -7 to -2 dBm.

3.7.8.6 Power feed and metering. (See 4.7.8.6) The output dc voltage shall be 800 Vdc maximum and the dc current shall be between 38 and 42 mA as indicated on the equipment display.

3.7.9 Full duplex signal detection and dynamic range. (See 4.7.9) The equipment receiver shall be capable of recovering data in a full-duplex circuit from a conditioned diphase signal transmitted over 0, 0.4, 0.8, 1.2, and 1.6 km of CX-11230 cable with or without power feed, with no errors and with the maintenance orderwire signal present in both directions. The maintenance orderwire signal shall be a maximum level of 18 Vrms at 1600 Hz. This performance shall be achieved with the receiver signal level varied +0, -4 dB about the nominal level at the maximum cable length.

3.7.10 Environmental operational test. (See 4.7.10) Environmental tests shall be comprised of full duplex signal detection at 2040 kb/s with the ring signal present and power feed operating. Proper operation shall be determined in accordance with 3.7.8.

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3.8 Service conditions. (See 4.8) The operational test of 3.7.10 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand exposure (non-operating) to ambient air temperatures as high as +71°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effects of dust particles that may accumulate within the shelter assembly or vehicle.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

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3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7). The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

3.9 Electromagnetic interference (EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4.

CE01	CS01	RE02	RS02
CE02	CS02	RE02.1	RS03
CE03	CS06		
CE04			
CE05			

NOTES:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The unit shall have a specified Mean-Time-Between-Failure (MTBF), θ_o of 6000 hours when in the normal modes of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

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3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements of their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA	DRAWING
BITE 1	*SM-D-875948
Orderwire Transmitter	SM-D-875979
Orderwire Receiver	SM-D-875980
Traffic Receiver A Low Speed	SM-D-875985
Traffic Transmitter Low Speed	SM-D-875986
Traffic Receiver B Low Speed	*SM-D-876265
Multi VDC	SM-D-884567
DC Input	SM-D-884575
AC Input	SM-D-884579
400/800 VDC	SM-D-884598
AC Only	SM-D-884678

*Drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational test. (See 4.7.8) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.8 after all other inspections and tests are completed to determine that the unit is operating prior to packaging.

3.14 System safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/ spark gaps, lenses).

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3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001 with the exception of paragraphs 3.3.1.1 (BLAST) and 3.3.1.2 (THERMAL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

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c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete equipment. The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table I and shall be performed in the order specified therein except that all hardwired assemblies shall undergo inprocess Group A inspection prior to burn-in.

4.3.3 First article inspection data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I. First article inspection.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by Subsidiary Documents	N/A	4.4	Inspections to be performed on all units.												
2. Burn-In	3.6	4.6	Inspections to be performed on all units.												
3. Group A inspection	(See Table II)		Inspections to be performed on all units.												
4. Group B Inspection	(See Table III)		Inspections to be performed on all units.												
5. Group C Inspection															
Subgroup 1															
High Temperature	3.8.1.1	4.8.1.1	1	*											
Low Temperature	3.8.1.2	4.8.1.2	2	*											
Subgroup 2															
EMI	3.9	4.9			1										
Vibration/Shock	3.8.7	4.8.7				1	*								
Subgroup 3															
Humidity (Note c)	3.8.2	4.8.2						1			*				
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust (Note c)	3.8.4	4.8.4							2		*				
Salt Fog (Note c)	3.8.5	4.8.5							3						
Fungus (Note c)	3.8.6	4.8.6								1					
Subgroup 5															
Signal Balance	3.7.1.2	4.7.1.2										1			
Clock/Data Phasing	3.7.1.4	4.7.1.4										2			
Offset	3.7.1.5	4.7.1.5**										3			
Slew Rate	3.7.1.6	4.7.1.6										4			

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TABLE I. First article inspection (Contd).

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
Input Load Impedance	3.7.1.8	4.7.1.8										5			
Clock Asymmetry	3.7.1.9	4.7.1.9										6			
Receiver Input Impedance	3.7.2.3	4.7.2.3										7			
Overshoot, Undershoot and Droop	3.7.2.4	4.7.2.4										8			
Line-to-Ground Voltage	3.7.1.3	4.7.1.3**										9			
Unloaded Output	3.7.1.10	4.7.1.10										10			
Dimensional Interchange	3.11	4.11										11			
Frame Acquisition Time	3.7.5	4.7.5										12			
Input Impedance	3.7.4.2	4.7.4.2										13			
Output Impedance	3.7.4.3	4.7.4.3										14			
Rise and Fall Times	3.7.2.2	4.7.2.2										15			
Input Impedance	3.7.3.1	4.7.3.1										16			
Output Impedance	3.7.3.3	4.7.3.3										17			
Signal-to-Distortion Ratio	3.7.3.2	4.7.3.2**										18			
Frequency Response	3.7.3.4	4.7.3.4**										19			
Power Variation	3.7.6.1	4.7.6.1										20			
Current Protection	3.7.6.2	4.7.6.2.2**										21	*		
Fault Locate	3.7.6.3	4.7.6.3										22			
Full Duplex Signal Detection and Dynamic Range	3.7.9	4.7.9										23	*		
Safety	3.7.6.4	4.7.6.4										24			
6. Group D Inspection															
Reliability	3.10	4.10												8 units	

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NOTES for table I:

a. The Inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by Inspections 5 and 6 specify the order of inspections for the indicated unit. An asterisk (*) indicates the use of an auxiliary MD-1023(P)/G unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. The quantity of units shall be specified in the contract or purchase order. All first article units shall be subjected to Group A and Group B inspections.

e. A double asterisk (**) indicates a one-time engineering test to be covered by a separate engineering report, not part of the first article test report.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where Required
Printed wiring	3.2.2
Semiconductor devices and integrated circuits	3.2.3
Parts, materials, and processes	3.3
Finish	3.4
Marking	3.5

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4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table II shall be inspected for conformance to the inspections specified in table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies, and units (end equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares which are not installed into complete units, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of a. above shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and a 100% end equipment visual/mechanical and electrical inspection in accordance with table II, in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.

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TABLE II. Group A inspection.

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
Traffic Receiver A Low Speed	3.15	4.13	2.5%	4.0%
Traffic Receiver B Low Speed	3.15	4.13	2.5%	4.0%
Traffic Transmitter Low Speed	3.15	4.13	2.5%	4.0%
Orderwire Receiver	3.15	4.13	2.5%	4.0%
Orderwire Transmitter	3.15	4.13	2.5%	4.0%
Multi VDC	3.15	4.13	2.5%	4.0%
AC Input	3.15	4.13	2.5%	4.0%
400/800 VDC	3.15	4.13	2.5%	4.0%
BITE 1	3.15	4.13	2.5%	4.0%
AC Only	3.15	4.13	2.5%	4.0%
DC Input	3.15	4.13	2.5%	4.0%
Case and Covers	3.15	4.13	2.5%	10.0%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
Traffic Receiver A Low Speed	3.12	4.12	1.0%	*
Traffic Receiver B Low Speed	3.12	4.12	1.0%	*
Traffic Transmitter Low Speed	3.12	4.12	1.0%	*
Orderwire Receiver	3.12	4.12	1.0%	*
Orderwire Transmitter	3.12	4.12	1.0%	*
Multi VDC	3.12	4.12	1.0%	*
AC Input	3.12	4.12	1.0%	*
400/800 VDC	3.12	4.12	1.0%	*
BITE 1	3.12	4.12	1.0%	*
AC Only	3.12	4.12	1.0%	*
DC Input	3.12	4.12	1.0%	*
ELECTRICAL				
Operational Tests	3.7.8	4.7.8	1.0%	*
BITE and Alarms	3.7.7	4.7.7	1.0%	*

*All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in table II and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in table III.

TABLE III. Group B inspection.

Inspection	Reqt Para	Insp Para	AQL
Line-to-Line Voltage	3.7.1.1	4.7.1.1	4.0%
Receiver Dynamic Range	3.7.1.7	4.7.1.7	4.0%
Output Voltage Level	3.7.2.1	4.7.2.1	4.0%
Output Power Level	3.7.4.1	4.7.4.1	4.0%
Power Feed (Constant Current Check)	3.7.6.2	4.7.6.2.1	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units from lots that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing, in that order. The unit selected every eighth month shall be subjected to EMI testing.

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4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to altitude, dust and salt fog testing, and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table IV.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

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TABLE IV. Group C inspection.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High temperature	3.8.1.1	4.8.1.1
Low temperature	3.8.1.2	4.8.1.2
Subgroup 2		
Vibration/Shock	3.8.7	4.8.7
EMI	3.9	4.9
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5		
Clock/data phasing	3.7.1.4	4.7.1.4
Signal balance	3.7.1.2	4.7.1.2
Slew rate	3.7.1.6	4.7.1.6
Input load impedance	3.7.1.8	4.7.1.8
Clock asymmetry	3.7.1.9	4.7.1.9
Unloaded output	3.7.1.10	4.7.1.10
Receiver input impedance	3.7.2.3	4.7.2.3
Overshoot, undershoot and droop	3.7.2.4	4.7.2.4
Input impedance	3.7.3.1	4.7.3.1
Output impedance	3.7.3.3	4.7.3.3
Power variation	3.7.6.1	4.7.6.1
Safety	3.7.6.4	4.7.6.4
Full duplex signal detection and dynamic range	3.7.9	4.7.9
Dimensional interchangeability	3.11	4.11
Fault locate	3.7.6.3	4.7.6.3

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TABLE IV. Group C inspection (contd).

Inspection	Reqt Para	Insp Para
Frame acquisition time	3.7.5	4.7.5
Input impedance	3.7.4.2	4.7.4.2
Output impedance	3.7.4.3	4.7.4.3
Rise and fall times	3.7.2.2	4.7.2.2

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is reworked to specifications and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspection.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent Group D inspections, eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to Group D falures shall be as specified in the contract or purchase order.

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4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected as shown in figure 1. Test conditions of 4.10.3 shall apply except that the duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The requirements of 3.6 shall be met.

4.6.1 Burn-in procedure.

a. Set the Rate Switch of the unit under test (UUT) to position 11 (2048 kb/s).

b. Set the Group Simulator for 2048 kb/s operation.

c. Set the Digital Voice Orderwire/Data Orderwire (DVOW/DOW) Simulator #1 for 16 kb/s operation and DVOW/DOW Simulator #2 for 2 kb/s operation.

d. Connect DVOW/DOW Simulator #1 to the DVOW channel and DVOW/DOW Simulator #2 to the DOW channel.

e. Turn on the UUT power feed switch and the UUT power feed switch and monitor the S, SA and LS alarms.

4.7 Electrical tests. (See 3.7) Unless otherwise specified all tests shall be performed at 115 V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipments listed in table VI or their equivalents will be utilized during testing and will be calibrated in accordance with MIL-C-45662A. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless

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otherwise specified, special test equipment (see table VII and figure 2) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test (see table VIII). Table IX identifies the external interface connections with the test figures. Unless otherwise noted all resistors in test figures shall be .5 watts.

TABLE VI. Standard test equipment.

Description	Mfg./Model No. (or equivalent)
Electronic Counter	H.P. 5248L
Volt-Ohm Meter	Simpson 260
DC Milliameter	H.P. 412A
DC Voltmeter	H.P. 412A
Telephone Oscillator	H.P. 236A
Transmission Measuring Set	H.P. 3555B
Signal Generator	H.P. 606B
RF Voltmeter	H.P. 3406A
RMS Voltmeter	H.P. 603
RMS Voltmeter	H.P.3403A
Audio Oscillator	H.P. 200CD
Audio Oscillator	H.P. 201C
Wattmeter	Weston 432
Attenuator	H.P. 4436A
AC Voltmeter	H.P.403
Attenuator	Kay 3-30
Sampling Oscilloscope	Tektronix 7313
Oscilloscope	Tektronix 454
AC Power Source	ELGAR 501
Oscilloscope	Tektronix 7904
Selective Voltmeter	H.P. 3581C
DC Power Source	Sorensen DCR-150-5A

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TABLE VII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Group Simulator	A3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination.
Timing Source	A3051666	The Timing Source provides four independent group outputs, switch-selectable to positions between 72 and 4608 kb/s. Also provided are four independent Station Clock/ Telemetry Clock outputs, switch-selectable to 2, 16, or 32 kHz.
Timing Source (Alternate)		An alternate Timing Source is shown in figure 2 that may be used in lieu of the above.
DVOW/DOW Simulator	A3051669	The DVOW/DOW Simulator provides a Pseudo-random data generator and correlator at 2, 16, and 32 kb/s with a balanced signal interface. A TTL error output signal is provided for counting errors externally.

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TABLE VIII. Government Furnished Equipment (GFE)

Equipment	Description
CX-11230()/G Cable	Twin coax cable, 1.4 mile reels

TABLE IX. MD-1023(P)/G interface connections.

Connector	Function	PIN			Signal Type
		LO	HI	Gnd	
LINE SIGNAL OUT	LINE DATA OUT				DIPHASE
LINE SIGNAL IN	LINE DATA IN				DIPHASE
GROUP	GROUP DATA IN	A	B	M	Non-Return-To-Zero (NRZ)
	GROUP TIM IN	C	D	N	SQ WAVE
	GROUP DATA OUT	E	F	P	NRZ
	GROUP TIM OUT	G	H	R	SQ WAVE
	BLACK STA CLK OUT	J	K	S	
ORDERWIRE AND ALARM	DVOW DATA IN	A	B	R	NRZ
	DVOW DATA OUT	C	D		NRZ
	DVOW TIM OUT	E	F		SQ WAVE
	DOW DATA IN	G	H		NRZ
	DOW DAT OUT	J	K		NRZ
	RCV TIM OUT	L	M		SQ WAVE
	XMT TIM OUT	a	b		SQ WAVE
	ANALOG OW IN	N	P		BAL 600 OHMS
	ANALOG OW OUT	S	T		BAL 600 OHMS
	ALARM S	U	V		ALM=CLOSE CONT
	ALARM SA	W	X		ALM=CLOSE CONT
	ALARM LS	Y	Z		ALM=CLOSE CONT
*POWER	(-)LOW (+) HIGH	C	A	B	

* Polarity indicated for dc input power.

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4.7.1 Balanced equipment-side signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage levels. (See 3.7.1.1)

a. Connect the equipment as shown in figure 3.

b. Set the Group Simulator and UUT bit rate to 2048 kb/s and also set the Group Simulator to NO FRAME and RANDOM DATA.

c. For each of the following output signals measure the positive and negative voltage levels line-to-line (i.e., between high and low pins) using the oscilloscope circuit of figure 4.

Group Data Out
Group Timing Out
Black STA CLK Out
DVOW Data Out
DVOW Timing Out
DOW Data Out
DOW Rcv Timing Out
DOW Xmt Timing Out

d. Verify that the requirements of 3.7.1.1 are met.

4.7.1.2 Signal balance. (See 3.7.1.2)

a. The equipment set up is the same as in 4.7.1.1.

b. For each of the output signals given in 4.7.1.1 c, measure the positive and negative voltage levels line-to-line (i.e., between high and low pins) as in 4.7.1.1.

c. Compare the positive and negative values. Determine the larger of the two values.

d. Compare the difference between the positive and negative values to 10% of the larger value.

e. Verify that the difference is less than 10% of the larger value.

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4.7.1.3 Line-to-ground voltage. (See 3.7.1.3)

a. The equipment setup is the same as in 4.7.1.1 except that the scope input must be unbalanced (see figure 5).

b. For each of the signals given in 4.7.1.1 c, with the Group signal at a 2048 kb/s rate, measure the voltage peaks on the high and low pins to ground.

c. Verify that the requirements of 3.7.1.3 are met.

4.7.1.4 Clock data phasing. (See 3.7.1.4)

a. The equipment set up is the same as in 4.7.1.1 except connect the oscilloscope as shown in figure 6.

b. For each set of signals given below, measure the delay between the zero voltage points on both the data rise and fall time slopes to the zero voltage points of the timing rise slopes, as indicated in figure 7.

Signal	Delay
Group Data/Timing Out 2048 kb/ds	± 29 ns*
DVOW Data/Timing Out 16 kb/s	± 1500 ns*
DOW Data/RCV Timing 2 kb/s	± 1500 ns*

* Positive means timing leads data.

c. Verify that the requirements of 3.7.1.4 are met.

4.7.1.5 Offset. (See 3.7.1.5)

a. The equipment set up is the same as in 4.7.1.1 with the oscilloscope connected as shown in figure 8.

b. For each of the signals given in 4.7.1.1 c, with the Group Rate set to 2048 kb/s, measure the dc offset voltage.

c. Verify that the requirements of 3.7.1.5 are met.

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4.7.1.6 Slew rate. (See 3.7.1.6)

a. The equipment set up is the same as in 4.7.1.1 with the oscilloscope connected as shown in figure 9.

NOTE: All measurements shall be made through 100 feet of RG-108 cable.

b. For each of the signals given in 4.7.1.1 c, with the group rate at 2048 kb/s, measure the time between the +40 mV and -40 mV points around the zero voltage level of the rise and fall slopes as shown in figure 10.

c. Verify that the requirements of 3.7.1.6 are met.

4.7.1.7 Receiver dynamic range. (See 3.7.1.7)

a. Connect the equipment as shown in figure 11. A variable attenuator will be connected in the normal signal path during the course of the test.

b. Set the Group Simulator to 2048 kb/s. Connect the attenuators in each signal path. Using the oscilloscope, adjust each attenuator to give a ± 50 mV peak signal into the UUT.

c. Observe the Error Counter at each signal path. Verify that there are no DVOW or DOW errors over a 30 second period, and no Group errors over a 10 second period.

4.7.1.8 Input load impedance. (See 3.7.1.8)

a. Connect the equipment as shown in figure 12.

b. Connect the test setup in sequence to the signal inputs given below.

(1) Group Timing In

(2) Group Data In

(3) DVOW Data In

(4) DOW Data In

c. At each signal inpt, adjust the audio oscillator to a frequency of 1.0 kHz for a V1 output voltage of 1.0 volts. Then measure the V2 voltage.

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d. Calculate the input impedance of each input using the formula:

$$Z = 100 \frac{V2}{V1} \text{ ohms}$$

e. Verify that the requirements of 3.7.1.8 are met.

4.7.1.9 Clock asymmetry. (See 3.7.1.9)

a. The equipment test setup is the same as shown in 4.7.1.1.

b. For each of the signals given below, with the Group signal at 2048 kb/s, measure the time (A) between the rise and fall slopes at the zero voltage points. Then measure the time (B) between two successive rise slopes at the zero voltage points as indicated in figure 13.

Signal	Tolerance
Group Timing Out (2048 kb/s)	± 15 ns
STA CLK Out	± 15 ns
DVOW Timing Out	± 1500 ns
DOW RCV Timing Out	± 1500 ns
DOW XMT Timing Out	± 100 ns

c. Verify that time (A) is equal to one-half of time (B) within the requirements of 3.7.1.9.

4.7.1.10 Unloaded output. (See 3.7.1.10)

a. The equipment setup is the same as described in 4.7.1.1 except the 50 ohm termination resistors are removed.

b. For each of the signals given below, measure the positive and negative voltage levels line-to-line (i.e., between high and low pins).

DOW RCV Data Out

DOW RCV Timing Out

DOW XMT Timing Out

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c. Verify that the requirements of 3.7.1.10 are met.

4.7.2 Cable-side group signal characteristics. (See 3.7.2)

4.7.2.1 Cable-side output voltage level (diphase). (See 3.7.2.1)

a. Connect the equipment as shown in figure 14 with the Cable CUR switch off and data rate set to 2048 kb/s.

b. Measure the peak-to-peak amplitude of the output voltage, at Line Signal Out 1 of the UUT (see figure 15).

4.7.2.2 Rise and fall times. (See 3.7.2.2)

a. The equipment setup is the same as 4.7.2.1.

b. Measure the rise and fall times of the loaded output signal between the 10% and 90% points of the waveform slopes at the Line Signal Out connector on the UUT (see figure 15).

c. Verify that the requirements of 3.7.2.2 are met.

4.7.2.3 Receiver input impedance. (See 3.7.2.3)

a. Connect the equipment as shown in figure 16.

b. Tune the signal generator to 500 kHz and set its output level to 0.7 Vrms as measured by the V1 RF voltmeter.

c. Measure V2 voltage.

d. Repeat steps b and c with the signal generator tuned to 2304 kHz.

e. Calculate the input impedance at both the 500 kHz and the 2304 kHz frequency settings, using the formula:

$$Z = 58 \frac{V2}{V1-V2} \text{ ohms}$$

f. Verify that the calculated input impedance meets the requirements of 3.7.2.3.

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4.7.2.4 Overshoot, undershoot and droop. (See 3.7.2.4)

a. The equipment setup is the same as in 4.7.2.1 (figure 14).

b. Measure the overshoot, undershoot and droop of the loaded output signal at the Line Signal Out connector on the UUT as shown in figure 15.

c. Verify that the requirements of 3.7.2.4 are met.

4.7.3 Equipment-side maintenance orderwire (OW) signal characteristics. (See 3.7.3)

4.7.3.1 Input impedance. (See 3.7.3.1)

a. Connect the equipment to the Analog OW In as shown in figure 17.

b. Adjust the oscillator frequency to 1 kHz and set level V1 to 1.0 V.

c. Measure the V2 voltage.

d. Calculate the input impedance using the formula:

$$Z = 600 \frac{V2}{V1} \text{ ohms}$$

e. Verify that the requirements of 3.7.3.1 are met.

4.7.3.2 Signal-to-distortion ratio. (See 3.7.3.2)

a. Connect the equipment as shown in figure 18.

b. Connect the Audio Oscillator to the Analog OW In and set the Audio Oscillator to 1 kHz and adjust the level to -4 dBm.

c. Connect the Group Simulator to UUT Group connector and set the bit rate to 2048 kb/s.

d. Set the distortion analyzer meter to indicate 0 dBm.

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e. Measure the distortion at the Analog OW Out with the distortion analyzer tuned to 1 kHz.

f. Verify that the requirementrs of 3.7.3.2 are met.

4.7.3.3 Output impedance. (See 3.7.3.3)

a. Connect the equiment as shown in figure 19(A).

b. Connect the oscillator to the Analog OW In and adjust the oscillator frequency to 1 kHz and its output to -4 dBm.

c. Measure the Analog OW output open circuit voltage V1 at the OW connector on the UUT.

d. Connect the equipment as shown in figure 19(B).

e. With the oscillator connected as in step b measure the Analog OW loaded output voltage V2 at the OW connector.

f. Calculate the impedance using the formula:

$$Z = 600 \frac{V1-V2}{V2} \text{ ohms}$$

g. Verify that the requirements of 3.7.3.3 are met.

4.7.3.4 Frequency response (See 3.7.3.4)

a. Connect the equipment as shown in figure 20.

b. With the oscillator connected to the Analog OW In, set the oscillator frequency to 1000 Hz. Adjust the oscillator for an output of 977 mV at the Analog OW Out connector as read at V2. This establishes the reference output level.

c. Note the equivalent dB reference level at the input (V1).

d. While maintaining the reference input level (V1) adjust the oscillator frequency to 300 and 1700 Hz. At each frequency, read the Analog OW output at V2 and obtain its value in dB relative to the reference level established in step b.

e. Verify that the requirements of 3.7.3.4 are met.

NOTE: The 59 dB attenuator pad sets the AVOW level received from the cable interface low enough so that the automatic gain control (AGC) is inactive.

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4.7.4 Cable-side maintenance orderwire signal chacteristics. (See 3.7.4)

4.7.4.1 Output power level. (See 3.7.4.1)

a. Connect the equipment as shown in figure 21.

b. The selective voltmeter will filter out the data traffic.

c. Connect the oscillator to the Analog OW In and set the oscillator frequency to 1 kHz and its output level (V1) to 977 mV.

d. Measure the Analog OW loaded output voltage at the Line Signal Out connecter on the UUT.

e. Calculate the output power using the formula:

$$dBm = 20 \log V2 + 3.67$$

where

V2 is in Vrms.

f. Verify that the requirements of 3.7.4.1 are met.

4.7.4.2 Input impedance. (See 3.7.4.2)

a. Connect the equipment as shown in figure 22.

NOTE: The oscillator must float.

b. Set the oscillator output to 0.7 Vrms as measured at V1 and its frequency to 300 Hz.

c. Measure V2 voltage at the Line Signal In connector.

d. Compute the input impedance using the formula:

$$Z = 430 \frac{V2}{V1} \text{ ohms}$$

e. Verify that the requirements of 3.7.4.2 are met.

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f. Set the oscillator frequency to 3000 Hz and its output level to 0.7 Vrms and repeat steps c through e.

4.7.4.3 Output impedance . (See 3.7.4.3)

a. Connect the equipment as shown in figure 23. Disable the data traffic.

b. Connect the oscillator to the Analog OW In and set the oscillator frequency to 300 Hz and its output level to -10 dBm.

c. Measure the amplitude of the loaded output voltage (V2) and the open circuit output voltage (V1) at the Line Signal Out connector on the UUT.

d. Compute the output impedance Z using the formula:

$$Z = 430 \frac{V1-V2}{V2} \text{ ohms}$$

e. Verify that the requirements of 3.7.4.3 are met.

f. Repeat steps b through e for 3000 Hz.

4.7.5 Frame acquisition time. (See 3.7.5)

a. Connect the equipment as shown in figure 24. Connect the counter as shown to the appropriate test points on the UUT.

b. Set the Group Simulator and UUT group data rate to 2048 kb/s rate.

c. Set the Test Selector switch on the UUT to the Fault Locate position. Press the Fault Locate button on the UUT. (This will interrupt the data and cause the unit to lose frame).

d. Measure the time from resumption of data (negative signal at PIN 30 of OW XMT CCA) to frame-in (positive signal at PIN 17 of OW RCV CCA). Take 10 measurements and record the results.

e. Repeat steps c and d with the Group Simulator and UUT set to 72 kb/s.

f. Verify that the requirements of 3.7.5 are met.

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4.7.6 Prime power interface. (See 3.7.6)

4.7.6.1 Power variation. (See 3.7.6.1)

a. Connect the equipment as shown in figure 25.

b. For each of the combinations of voltage and frequency given below determine the input power and the Group bit error rate (BER).

NOTE: 28 Vdc test to be performed only on units that are equipped for ac/dc operations).

115 Vac	60 Hz
103.5 Vac	47.5 Hz 420 Hz
126.5 Vac	47.5 Hz 420 Hz
22.0 Vdc 28.0 Vdc 32.0 Vdc	

c. Verify the following for each case:

(1) Input power 150 watts maximum.

(2) No Group errors over a 10 second period.

4.7.6.2 Power feed. (See 3.7.6.2)

4.7.6.2.1 Power feed (constant current check).

a. Connect the equipment as shown in figure 26.

b. Measure the current of the Power Feed signal first with a shorted load, and then into the 20 kilohm load. Ensure that the milliammeter reads up-scale.

c. Verify that the Power Feed current is 40 mA dc $\pm$5% and that the polarity is correct.

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4.7.6.2.2 Current protection.

a. Connect the equipment as shown in figure 27, connect the Auxiliary Unit Line Signal Out connector to the UUT Line Signal Out connector.

b. Turn on the ac power to both units. Then turn on the Auxiliary Unit power feed voltage switch for one (1) minute. Then turn off the power feed voltage switch.

c. Repeat step b with the Auxiliary Unit Line Signal Out connected to the UUT Line Signal In connector.

d. Perform the operational test of 4.7.8 on the UUT.

e. Verify that the UUT sucessfully completes the operational test.

4.7.6.3 Fault locate. (See 3.7.6.3)

a. Connect the equipment as shown in figure 28, starting with the UUT connected to one TD-1218/G pulse restorer.

b. Turn on the CABLE CURRENT switch.

c. Set the CABLE TEST SELECT switch to FAULT LCT and press the PUSH TO FLT LCT push button.

d. Verify that the UUT displays 2 on its front panel display.

e. Add a second TD-1218 through 1 mile of CX-11230 cable and repeat step c. Verify that the UUT displays 3.

f. Add a third TD-1218 through 1 mile of CX-11230 cable and repeat step c. Verify that the UUT displays 4.

4.7.6.4 Safety. (See 3.7.6.4)

a. Connect the equipment as shown in figure 29.

b. Short the power feed signal to ground for 5 seconds. After removing the short, verify that the power feed current returns to 40 mA nominal.

c. Measure the open circuit voltage (V_{OC}) 2 seconds after opening the load switch.

d. Verify that the open circuit voltage is less than 30 volts after 2 seconds.

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4.7.7 BITE and alarms. (See 3.7.7)

4.7.7.1 Alarms. (See 3.7.7.1)

a. Connect the equipment as shown in figure 30. Set the Group Simulator and UUT to rate 11 (2048 kb/s) and the cable current to OFF.

b. No fault condition.

(1) Verify, using an ohmmeter, that the following BITE conditions are met:

LS OPEN

S OPEN

SA OPEN

SUMMARY FAULT OFF

All LEDs OFF

(2) Remove the BITE CCA and verify that no Group, DOW and DVOW errors are generated.

(3) Reinsert the BITE CCA and verify that no Group, DOW or DVOW errors are generated.

c. Loss of input.

(1) Disconnect the CX-11230 cable from the Line Signal Out connector on the UUT.

(2) Verify that the following BITE requirements are met.

S CLOSED

LS CLOSED

SA OPEN

SUMMARY FAULT lamp illuminates.

LED on TRF RCV A CCA illuminates.

(3) Reconnect the CX-11230 cable to the Line Signal Out connector on the UUT.

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d. Alarm check operation.

(1) Press the LAMP TEST switch on the UUT.

(2) Verify that the following BITE requirements are met:

S CLOSED

LS CLOSED

SA CLOSED

SUMMARY FAULT lamp and Power Out of Tolerance (POT) lamp illuminates. Verify that no Group, DOW and DVOW errors are generated.

e. Power loss.

(1) Turn off power at the source.

(2) Verify that the following BITE requirements are met:

S CLOSED

SA CLOSED

LS OPEN

f. Power off.

(1) Turn off the UUT power switch.

(2) Verify that the following BITE requirements are met:

S CLOSED

SA OPEN

LS OPEN

4.7.8 Operational rate test. (See 3.7.8)

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4.7.8.1 Group data error check with ring tone. (See 3.7.8.1)

a. The equipment set up is the same as in 4.7.7.1. Set the oscillator to 1600 Hz and its output to 977 mV (-4 dBm into the unit).

b. Set the Group Simulator to each of the rates as shown below. At each rate, monitor the Group Error Detector for a 10 second period. Verify that no errors are generated.

Group Rate, kb/s
72
128
512
576
1536
2048

4.7.8.2 Digital voice orderwire check. (See 3.7.8.2)

a. Monitor the 16 kb/s DVOW Error Detector for a 30 second period.

b. Verify that there are no errors generated.

4.7.8.3 Data orderwire check. (See 3.7.8.3)

a. Monitor the 2 kb/s DOW Error Detector for a 30 second period.

b. Verify that there are no errors generated.

4.7.8.4 Alarm monitor check. (See 3.7.8.4)

a. Monitor the S, SA and LS alarm outputs. Verify that there are no alarms generated, that is, there is an open circuit indicated on the ohmmeter. (A short indicates alarm).

b. Disconnect the Group cable on the Auxiliary Unit and verify that the S and SA alarm are shorted on the UUT.

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c. Reconnect the Group cable to the Group connector on the Auxiliary Unit. Disconnect the Line Signal Out cable on the Auxiliary Unit. Verify that the LS alarm is generated on the UUT.

4.7.8.5 Maintenance orderwire output level check. (See 3.7.8.5)

a. With the oscillator connected to the AVOW In and set to 1600 Hz at 977 mV output (-4 dBm into the unit), measure the AVOW output.

b. Verify that the output is between -7 and -2 dBm.

4.7.8.6 Power feed and metering check. (See 3.7.8.6)

a. Set up the equipment as shown in figure 26.

b. Turn on the power feed switch on the UUT.

c. Measure the power feed voltage and current.

d. Using the UUT meter, measure the transmit current, receive current and voltage.

e. Verify that the UUT current and voltage measurements are within ±5% of the measurement of step c.

f. Turn off the power feed switch on the UUT.

4.7.9 Full-duplex signal detection and dynamic range. (See 3.7.9)

a. Connect the equipment as shown in figure 31.

b. Set the Group Simulator for 2048 kb/s.

c. Set the DVOW/DOW Simulators for 16 kb/s and 2 kb/s, respectively.

d. Set the oscillator to 1600 Hz and its output level to -4 dBm.

e. Monitor the Group errors over a 10 second period. Verify that there are no errors.

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f. Monitor the DVOW and DOW Error Detectors over a 30 second period. Verify that there are no errors.

g. Remove the CX-11230 cable and the 4 dB attenuator (i.e., connect the equipments together using less than 10 feet of cable).

h. Repeat steps e and f.

4.7.10 Environmental operational test. (See 3.7.10)

a. Install the test items in the test facility and connect the equipment as shown in figure 32.

b. Set the Group Simulator, UTT and Auxiliary Unit Group rate to 2048 kb/s.

4.7.10.1 Power feed and metering. (See 3.7.8.6)

WARNING

AN OPEN CIRCUIT POTENTIAL OF 1000 VOLTS APPEARS ON THE CABLE AND ON THE LINE SIGNAL OUT CONNECTOR WHEN THE MD-1023(P)/G CABLE CURRENT SWITCH IS ON.

a. Turn on the CABLE CURRENT switch on the Auxiliary Unit and the UUT.

b. Measure the receive voltage and current at the Auxiliary Unit using the internal meter. The output voltge shall be 800 Vdc max and the current shall be 38 to 42 mA dc.

c. Turn off the CABLE CURRENT switch on the Auxiliary Unit and the UUT. Remove the load box from the circuit.

4.7.10.2 Group data error check with ring tone. (See 3.7.8.1)

a. Press and hold the RING button on the Auxiliary Unit.

b. Monitor the Group error detector for a period of 10 seconds.

c.Verify that there are no errors.

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4.7.10.3 Digital voice orderwire error check. (See 3.7.8.2)

a. Press and hold the RING button on the Auxiliary Unit.

b. Monitor the DVOW error detector for a period of 30 seconds.

c. Verify that there are no errors.

4.7.10.4 Data orderwire error check. (See 3.7.8.3)

a. Press and hold the RING button on the Auxiliary Unit.

b. Monitor the DOW errors for a period of 30 seconds.

c. Verify that there are no errors.

4.7.10.5 Alarm monitor check. (See 3.7.8.4)

a. Monitor the S, SA and LS alarm outputs. Verify that there are no alarms generated, that is, there is an open circuit indicated on the ohmmeter. (A short indicates alarm).

b. Disconnect the Group cable on the Auxiliary Unit and verify that the S and SA alarm are shorted on the UUT.

c. Reconnect the Group cable to the Group connector on the Auxiliary Unit. Disconnect the Line Signal Out cable on the Auxiliary Unit. Verify that the LS alarm is generated on the UUT.

4.7.10.6 Maintenance orderwire output level check. (See 3.7.8.5)

a. With the oscillator connected to the AVOW In and set to 1600 Hz at 977 mV output (-4 dBm into the unit), measure the AVOW output.

b. Verify that the output level is between -7 and -2 dBm.

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4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hours cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C(+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C(-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation. (Para. II- 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C(-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low pressure (altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

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4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 200 ±100 feet per minute. In step 5, the air temperature shall be adjusted to +63°(+145°F). The equipment shall be reoriented periodically to expose all its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C(+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axes designated on the schedules are the vehicle axes. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. Prior to and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

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4.8.7.2 Shock.

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance to EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article. First article reliability testing shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliablity test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Two Cable Drive Modems MD-1023(P)/G shall be operated back-to-back.

b. Duty cycle. Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24 hour cycle.

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c. Chamber temperature shall be 40 ±5°C (95°F to 113°F).

d. Vibration shall be 2.2 g ±10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliability test method. The detailed test procedures and conditions for performing the test outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor prepared, Government approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.8.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic units for mating with case.

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e. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective TRS drawings listed in 3.12.

4.13 Visual and mechanical inspection. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (System safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

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4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table X, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

TABLE X. Packing inspection provisions.

NO.	CHARACTERISTICS	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitation	Weigh & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.15 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. Test for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I though VII, coded as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED.

5.1.2 Level B. 2M-1-00-00-NR-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The MD-1023(P)/G is used in shelters and vans and will be mounted in an equipment case. It provides signal and power interface with repeatered coaxial cable systems via CX-11230 cable and provides transmission of analog voice, digital voice and data orderwires. The modem interfaces with multiplexers and pulse form restorers in the DGM family.

6.1.1 Functional description. The MD-1023(P)/G is designed to provide repeatered cable transmission for distances up to 40 miles (64 km) or unrepeatered cable transmission up to 1 mile (1.6 Km) in length. It accepts on its equipment side balanced NRZ group inputs which are transmitted to the cable as conditioned diphase, and provides the inverse function in the other direction. The unit accepts input bit rates ranging from 72 to 2048 kb/s on its equipment side, which are transmitted to the cable at a fixed rate of 2304 kb/s, and provides the inverse function in the other direction. The unit provides a 16 kb/s DVOW interface and a 2 kb/s DOW interface. These signals are Time Division Multiplexed (TDM) with the data to form the 2304 kb/s signal. The modem supplies the necessary power to operate Pulse Restorers TD-1218/G, as well as analog facilities for locating a faulty pulse restorer, accurate to ±1 restorer.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article inspection units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirement.

i. MIL-Q-9858

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of all first article tests.

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6.5 Location of operational inspection. (See 4.7.8) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VIII.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key-words) list.

Analog Voice Orderwire	DOW
AVOW	DVOW
Balanced	Group
Conditioned Diphase	Low Speed Cable Driver Modem (LSCDM)
Data Orderwire	Modem
Digital Group Multiplex	Modem, Cable Driver
Digital Voice Orderwire	Modem, Cable Driver, MD-1023(P)/G
Diphase	Orderwire

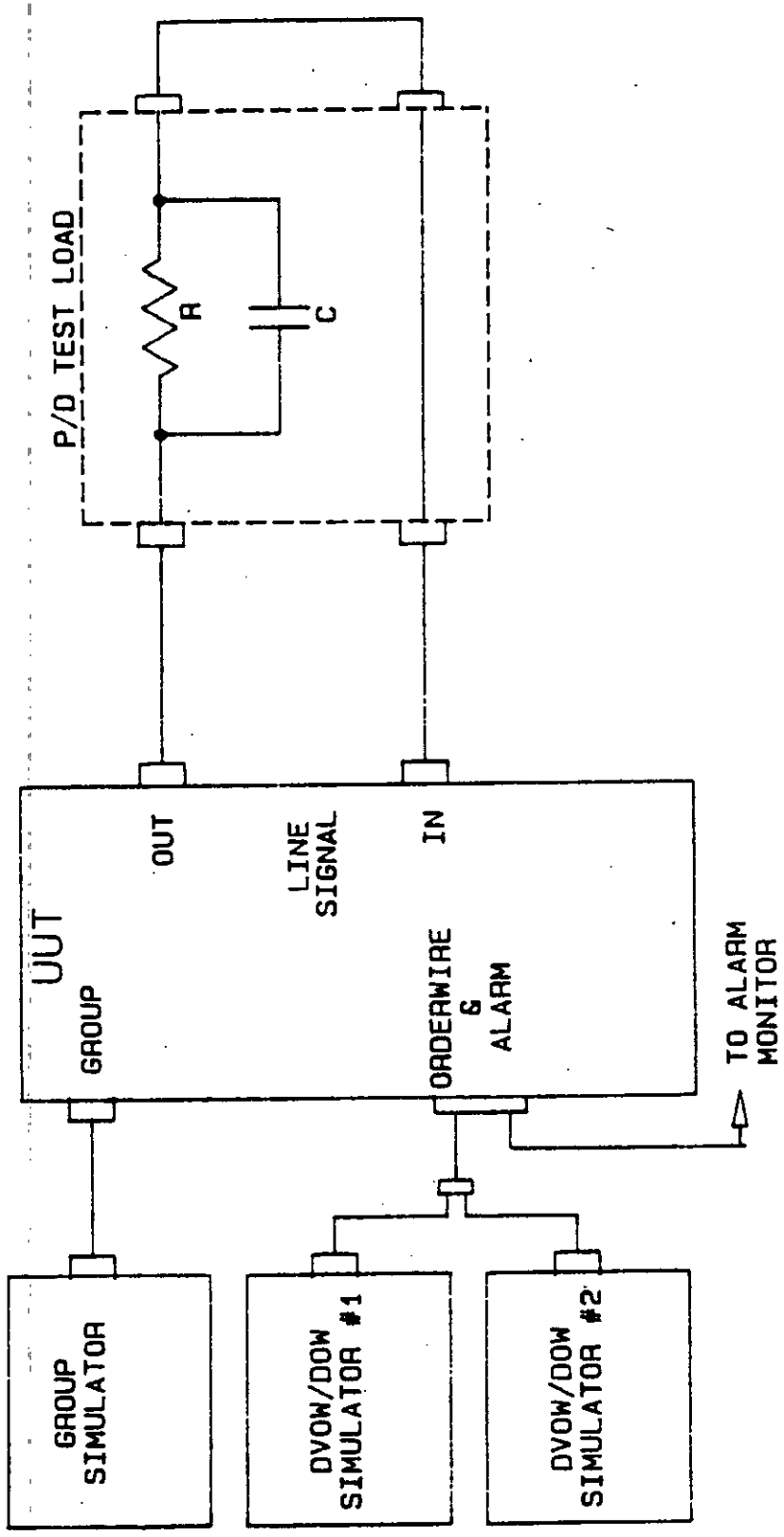
CUSTIODIAN:

ARMY-CR

PREPARING ACTIVITY

ARMY-CR
Project NO. 5820-A822

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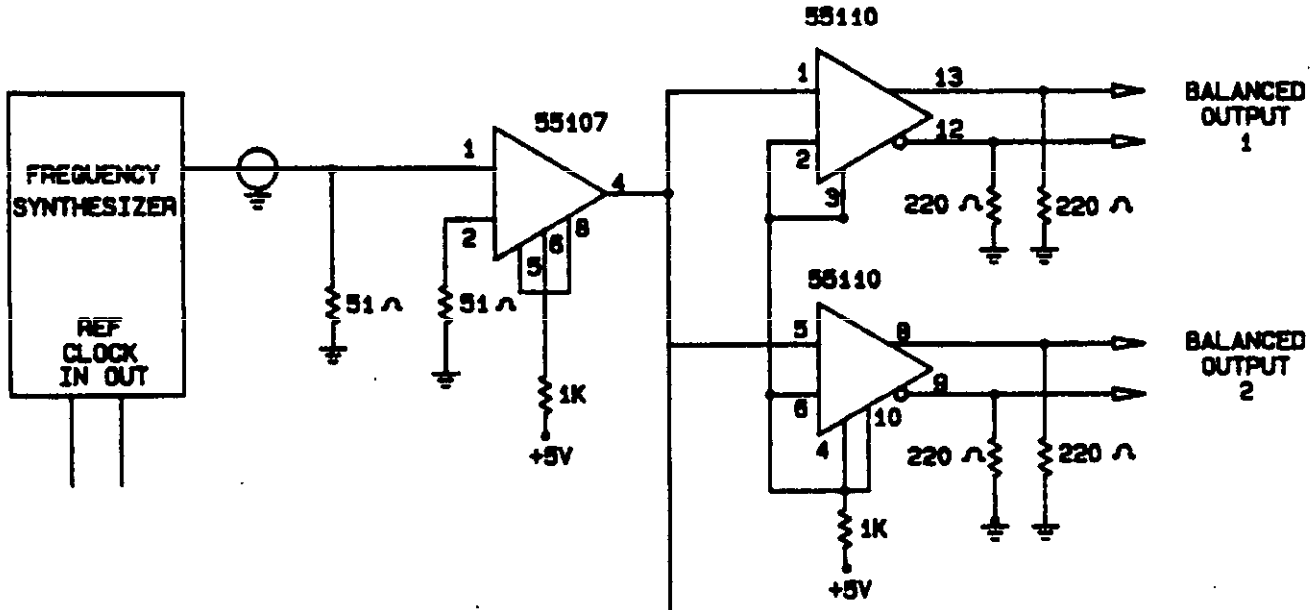


R 10K OHMS ±1%, 50W

C 1.0 uF, 1000V

FIGURE 1. Burn-in.

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GROUP RATE	SYNTHESIZER FREQUENCY
1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 2. Alternate timing source.

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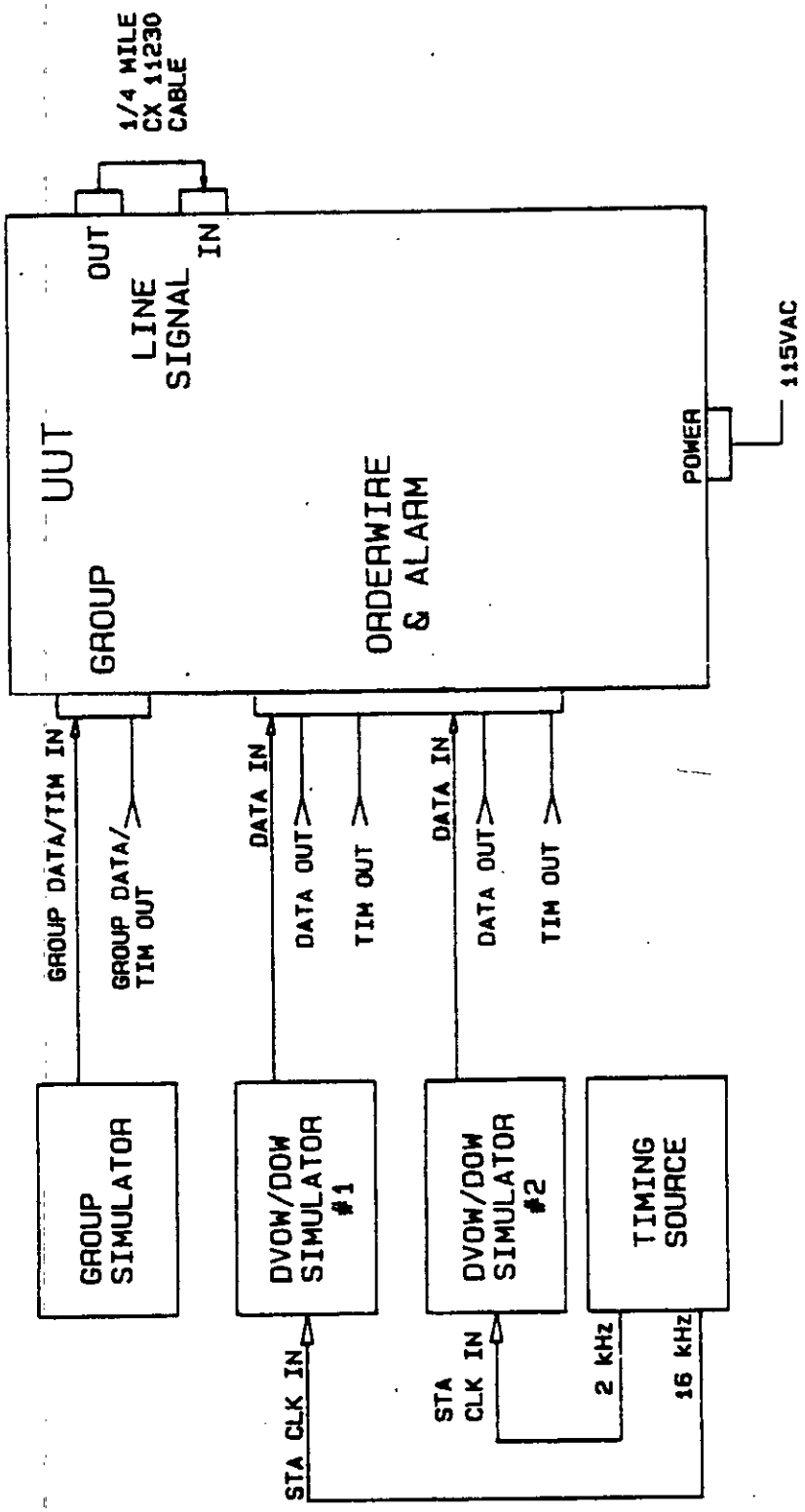
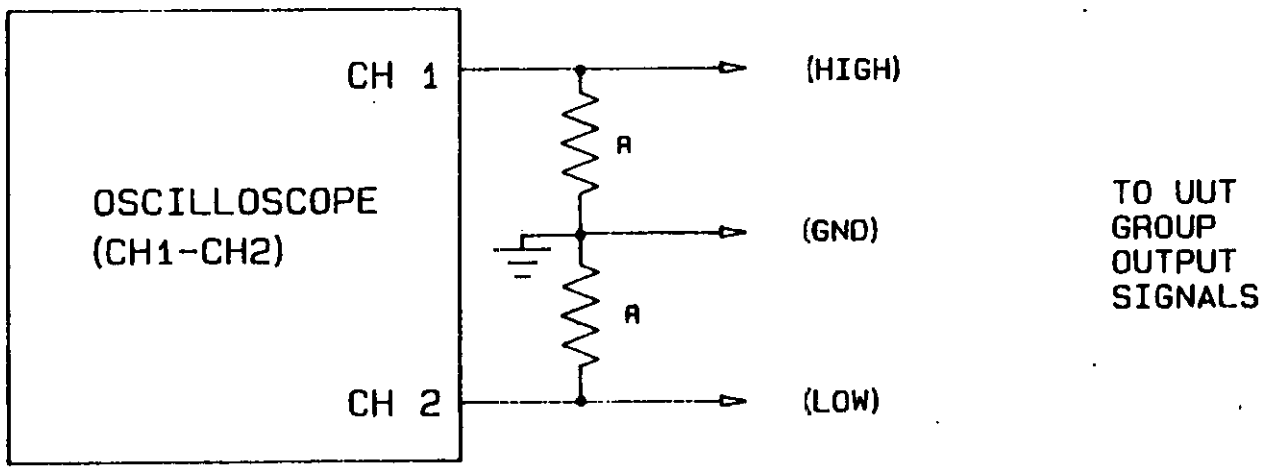


FIGURE 3. Line-to-line voltage.

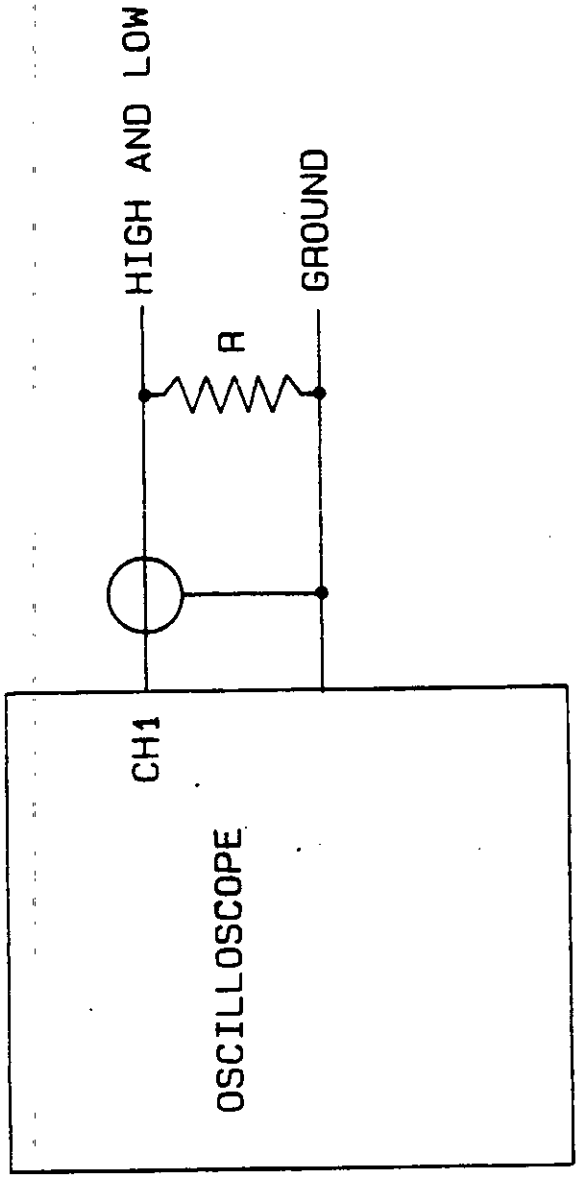
Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the currentversion before use.



R 50 OHMS ±1%

FIGURE 4. Oscilloscope configuration: Line-to-line voltage levels; signal balance.

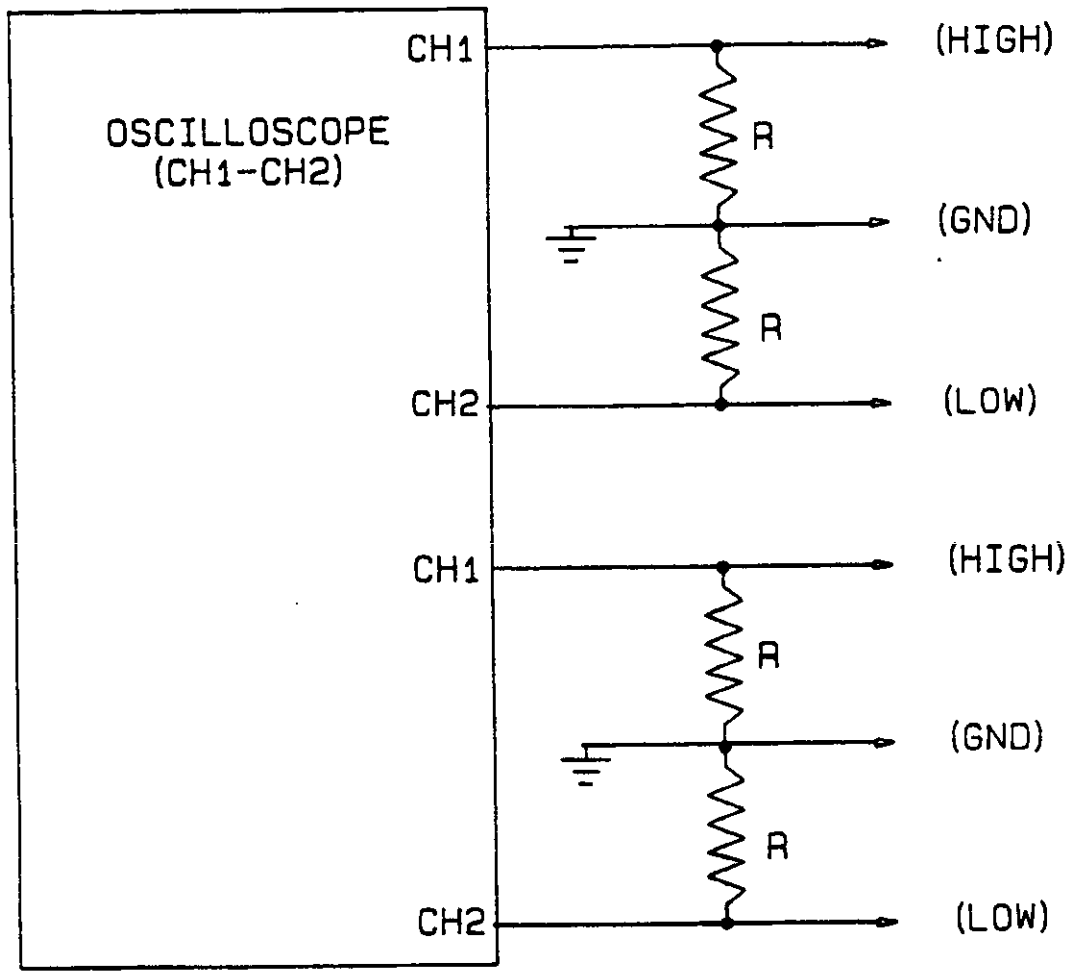
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
Check the source to verify that this is the current version before use.



R 50 OHMS ±1%

FIGURE 5. Oscilloscope configuration: Line to ground.

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R 50 OHMS ±1%

FIGURE 6. Oscilloscope configuration: Clock/data phasing.

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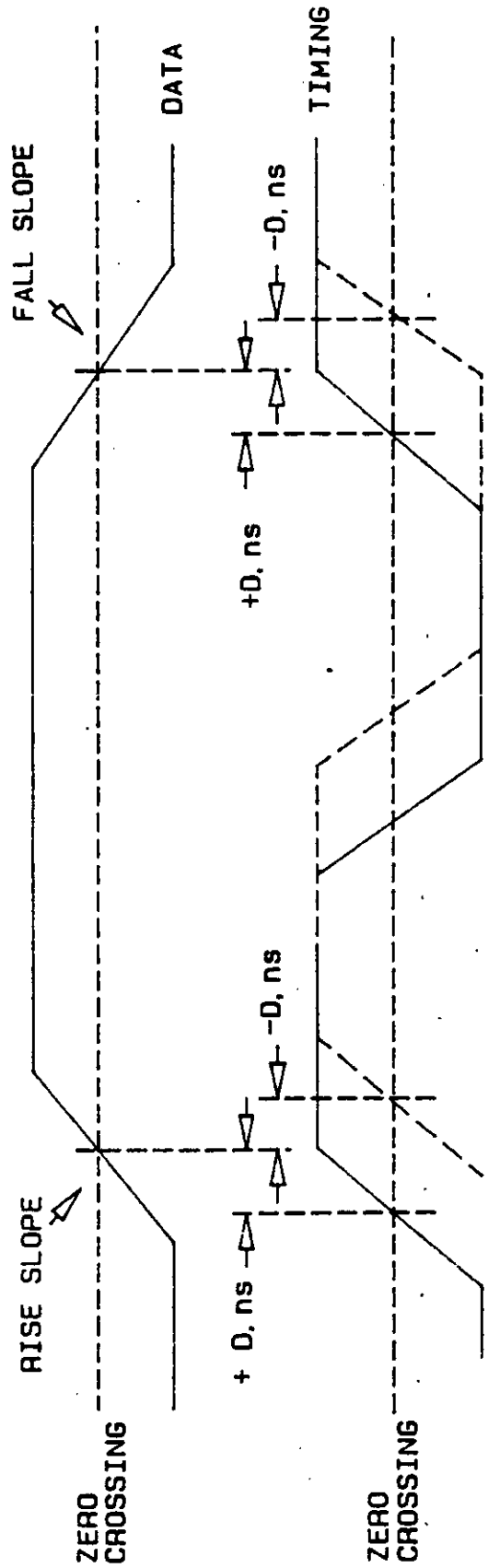
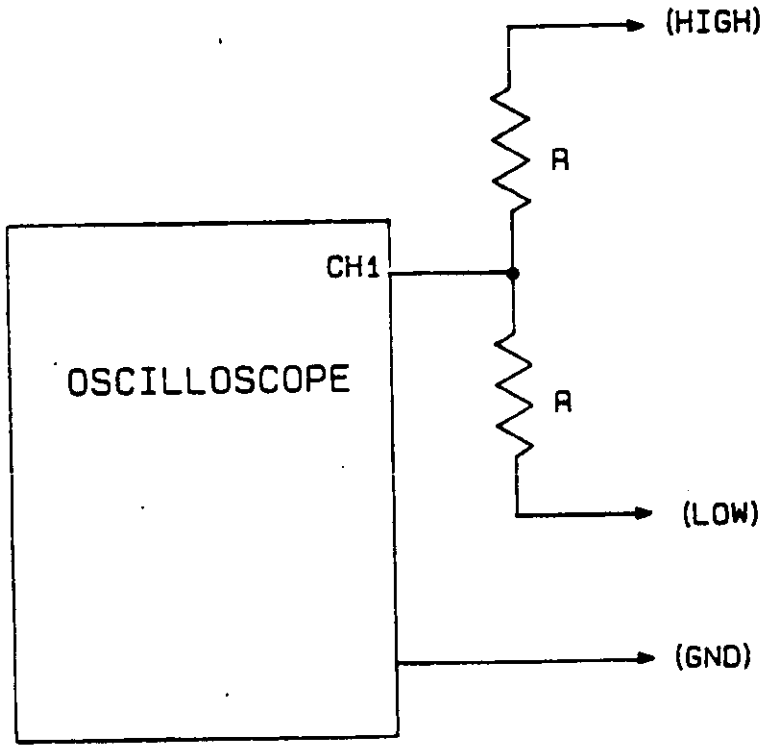


FIGURE 7. Delay measurement (clock/data phasing).

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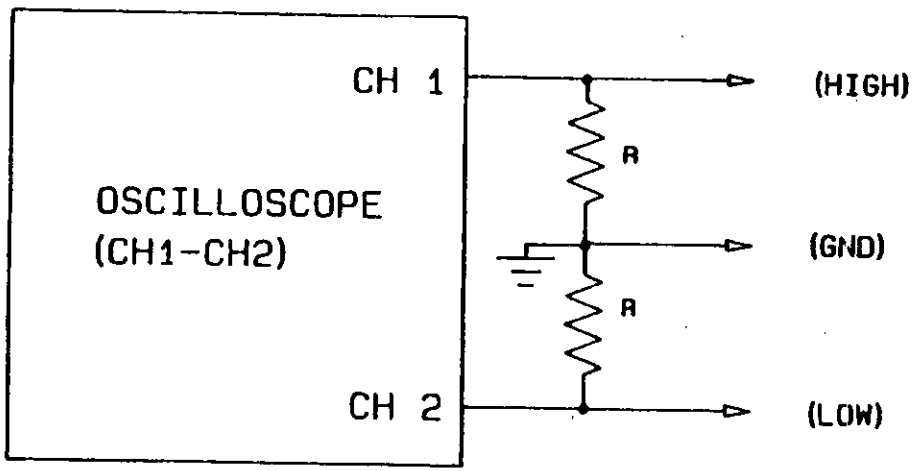


R 50 OHMS ±1%

FIGURE 8. Oscilloscope configuration: Offset.

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Source: https://assist.dla.mil -- Downloaded: 2026/10-26T03:59Z
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R 39 OHMS ±1%

FIGURE 9. Oscilloscope configuration: Slew rate.

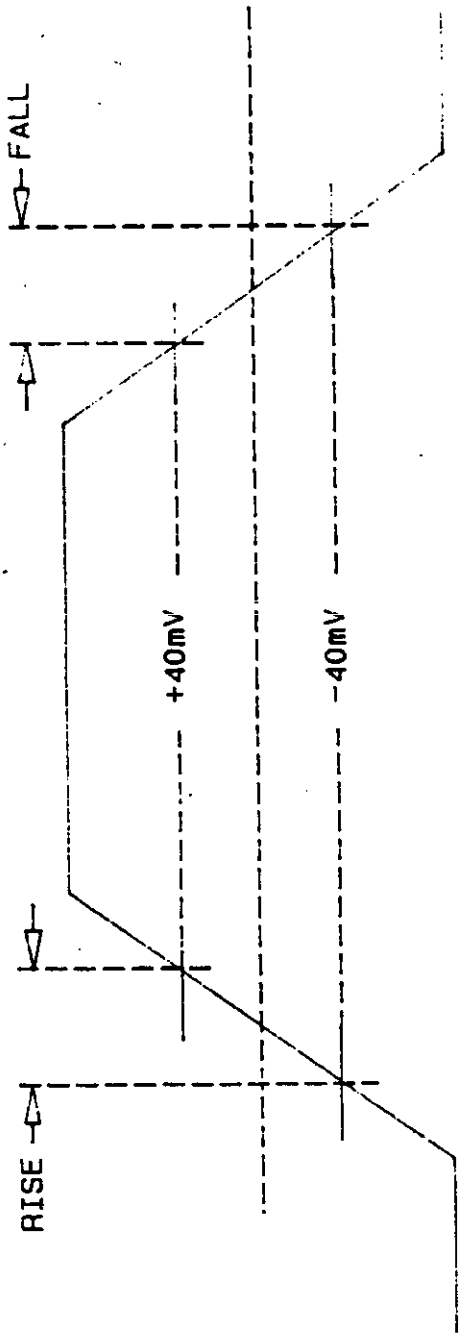


FIGURE 10. Rise and fall slopes (slew rate).

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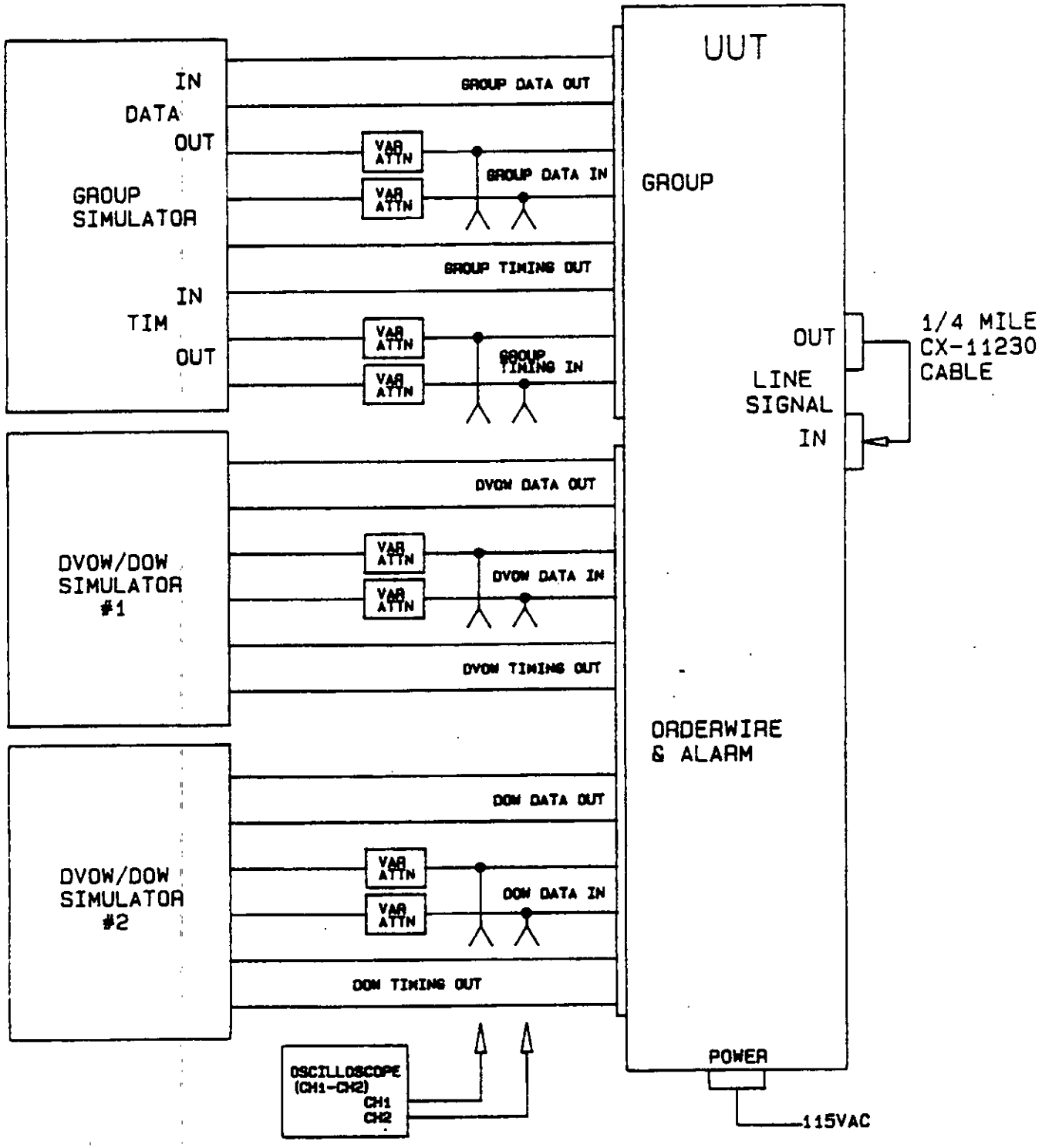


FIGURE 11. Receiver dynamic range.

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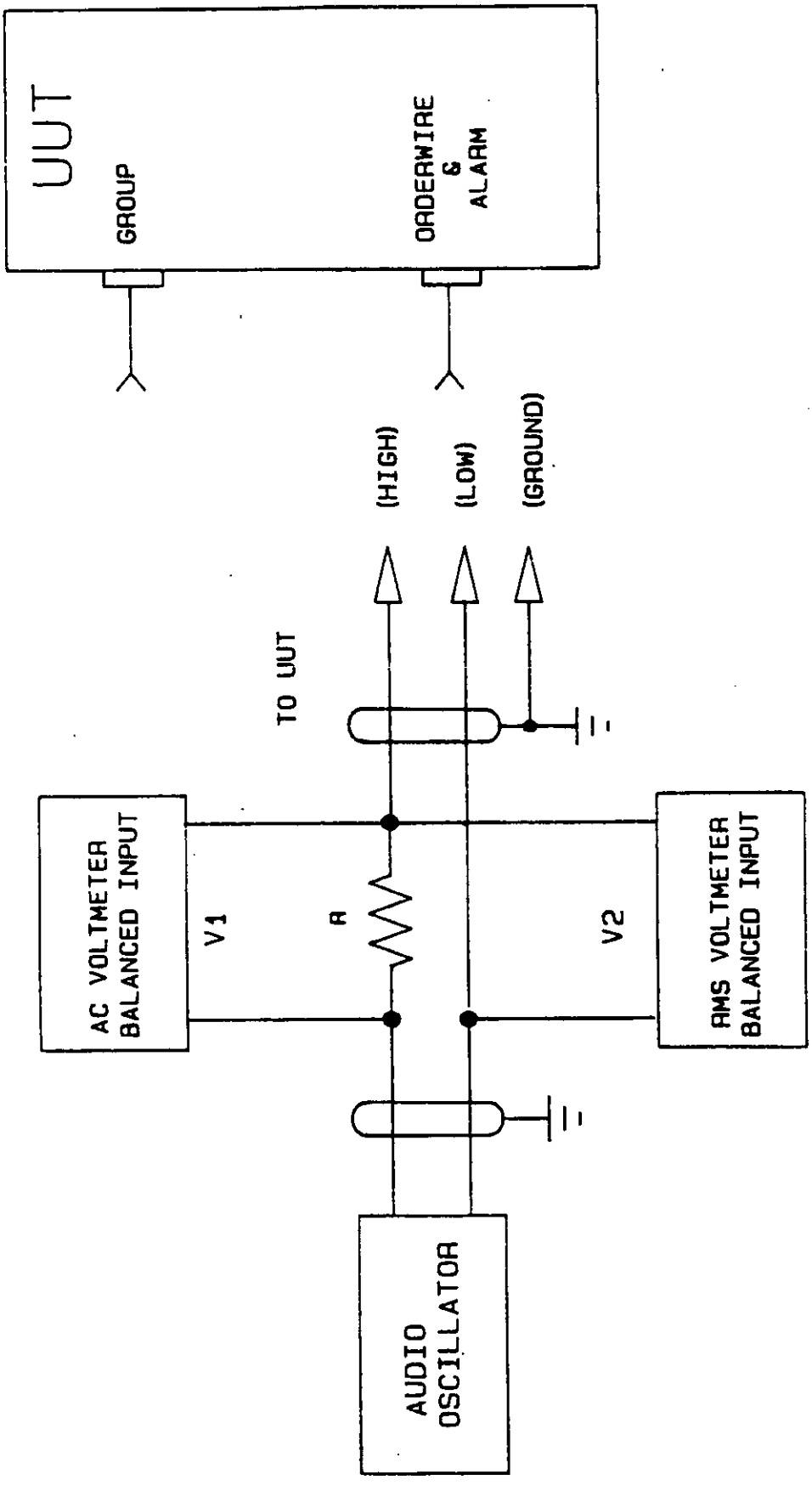


FIGURE 12. Input load impedance.

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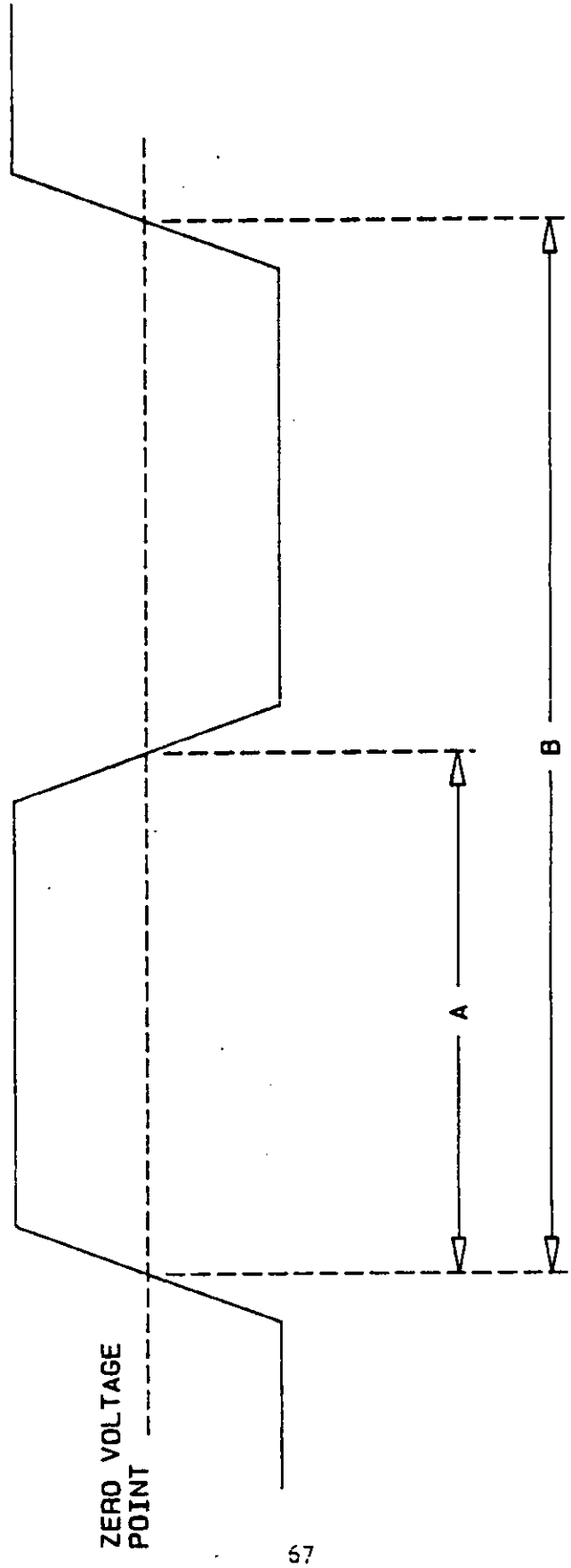
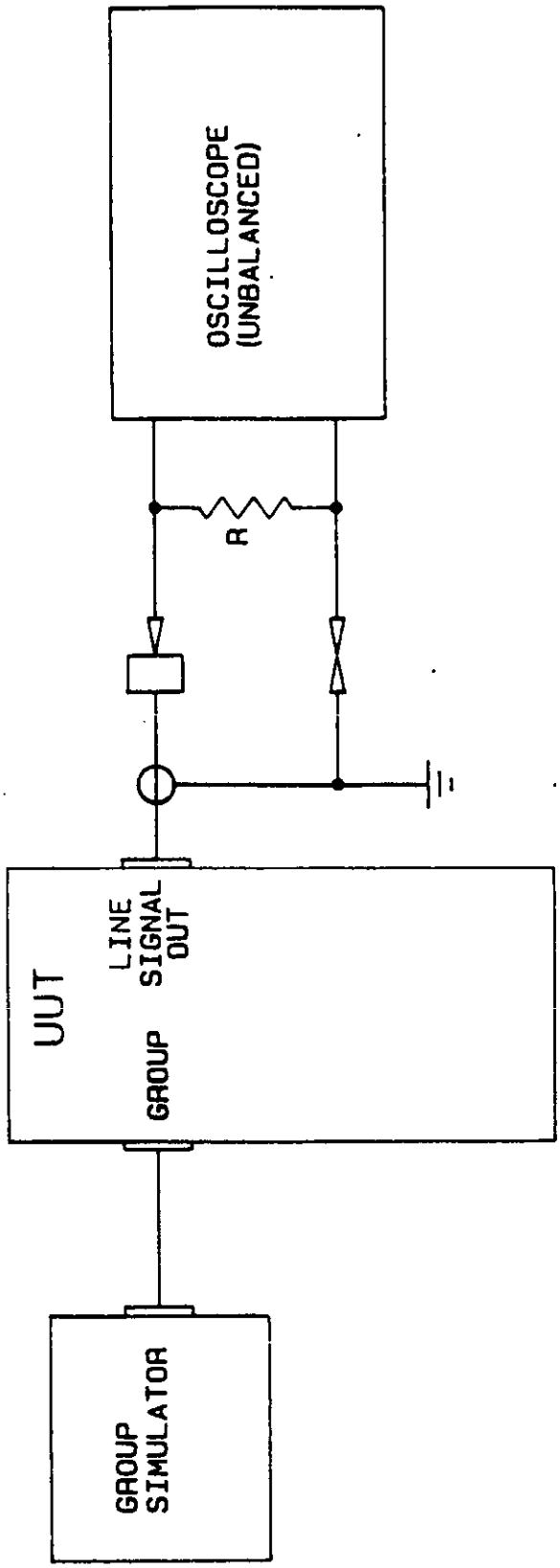


FIGURE 13. Clock asymmetry.

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R 58 OHMS ± 1%

FIGURE 14. Output voltage levels.

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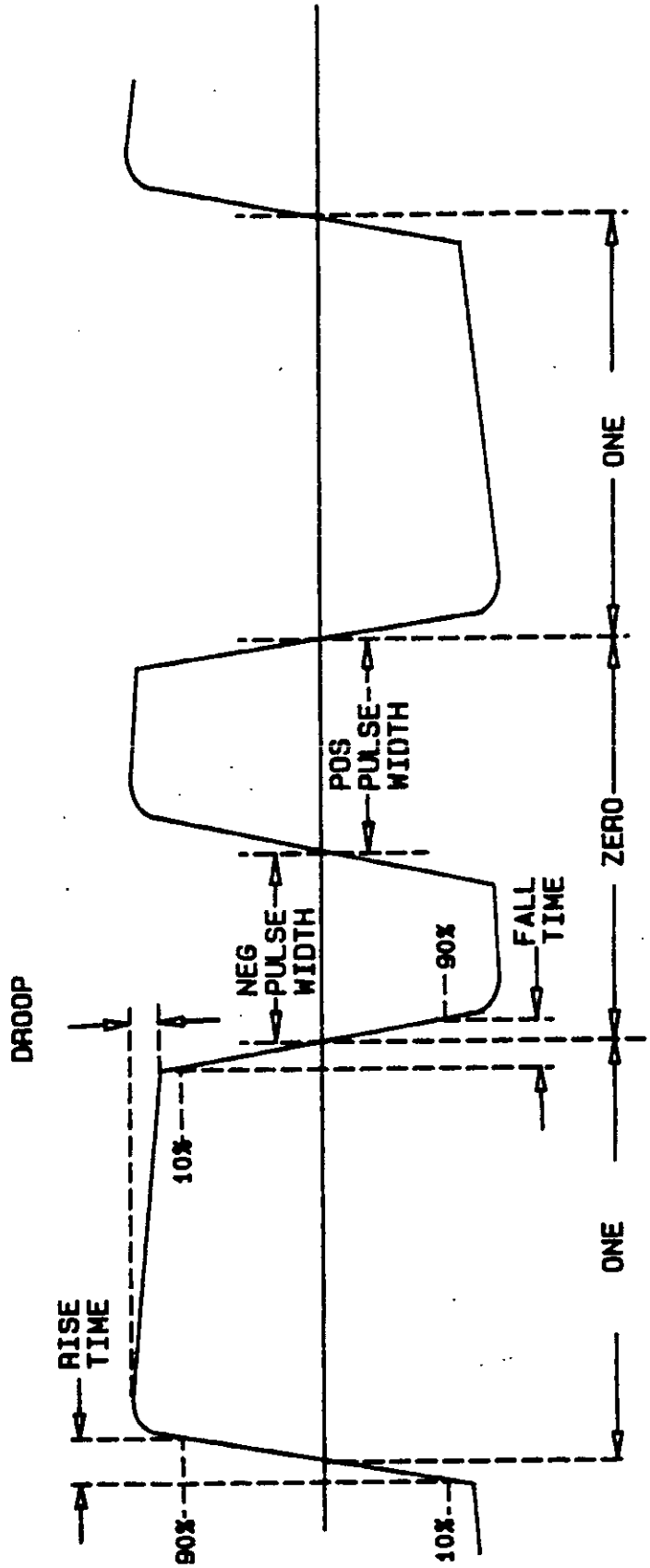


FIGURE 15. Diphase waveform (digital zero).

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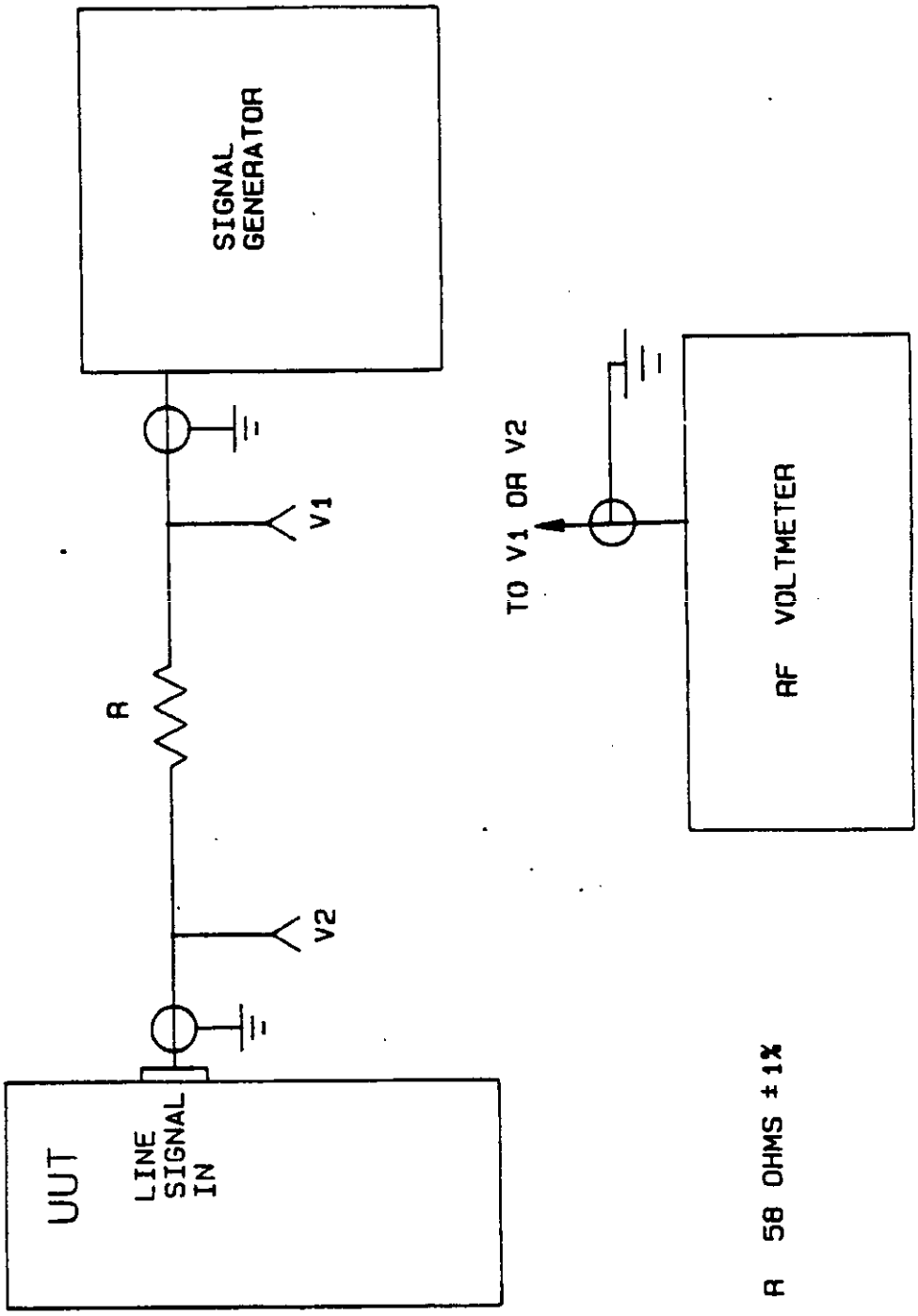
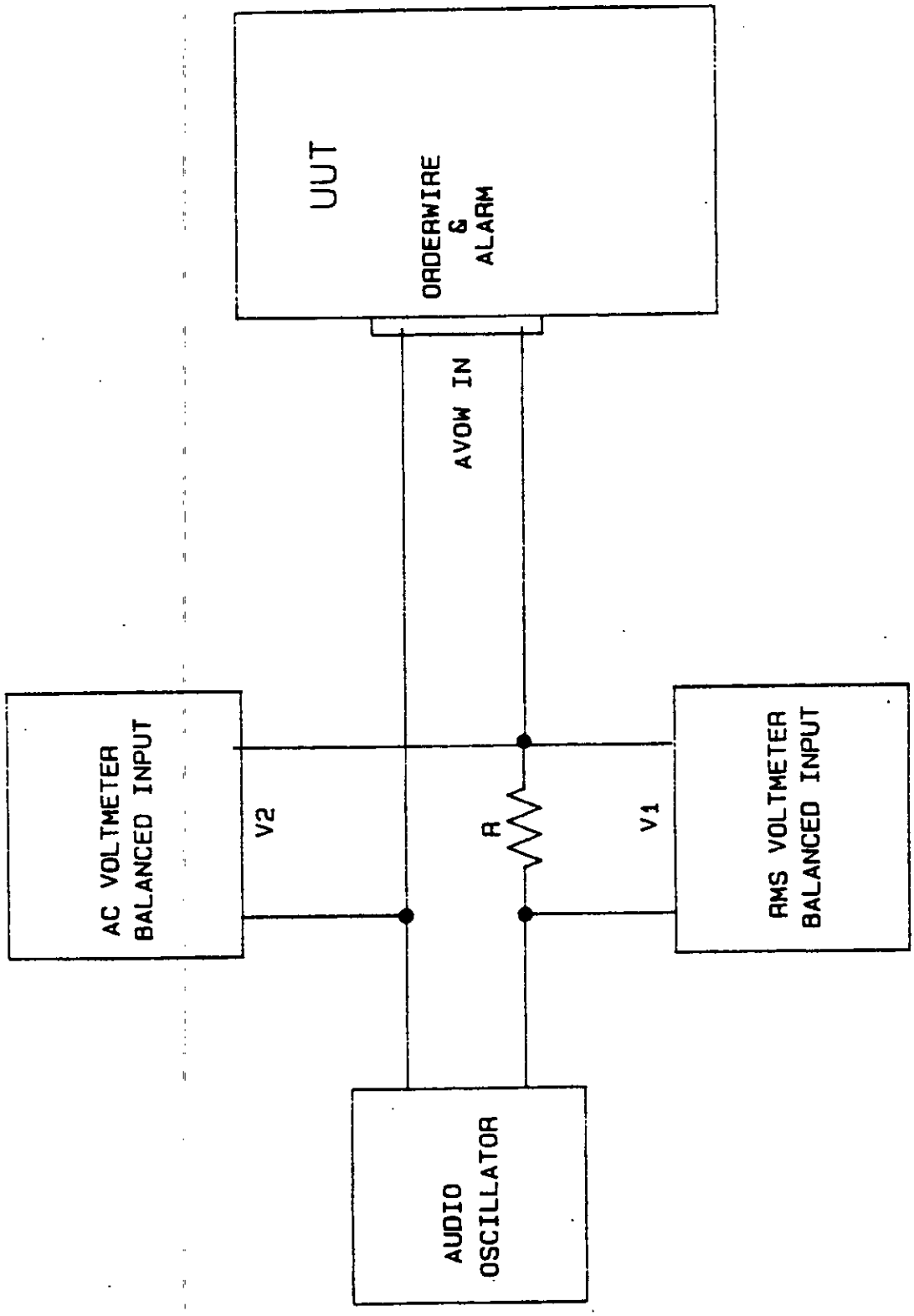


FIGURE 16. Receiver-input impedance.

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R 600 OHMS ±1%

FIGURE 17. Equipment-side maintenance orderwire input impedance.

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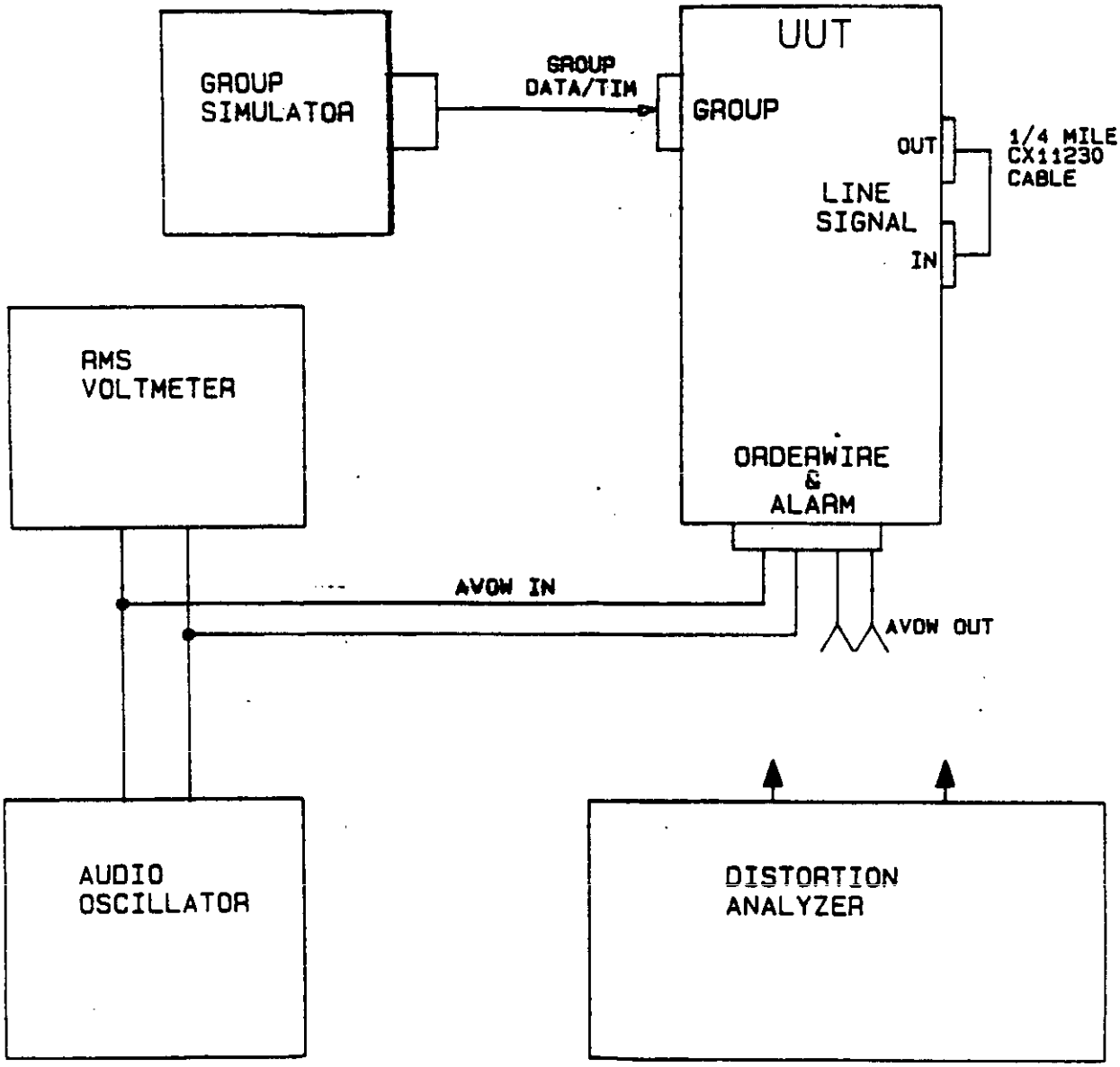


FIGURE 18. Signal-to-distortion ratio.

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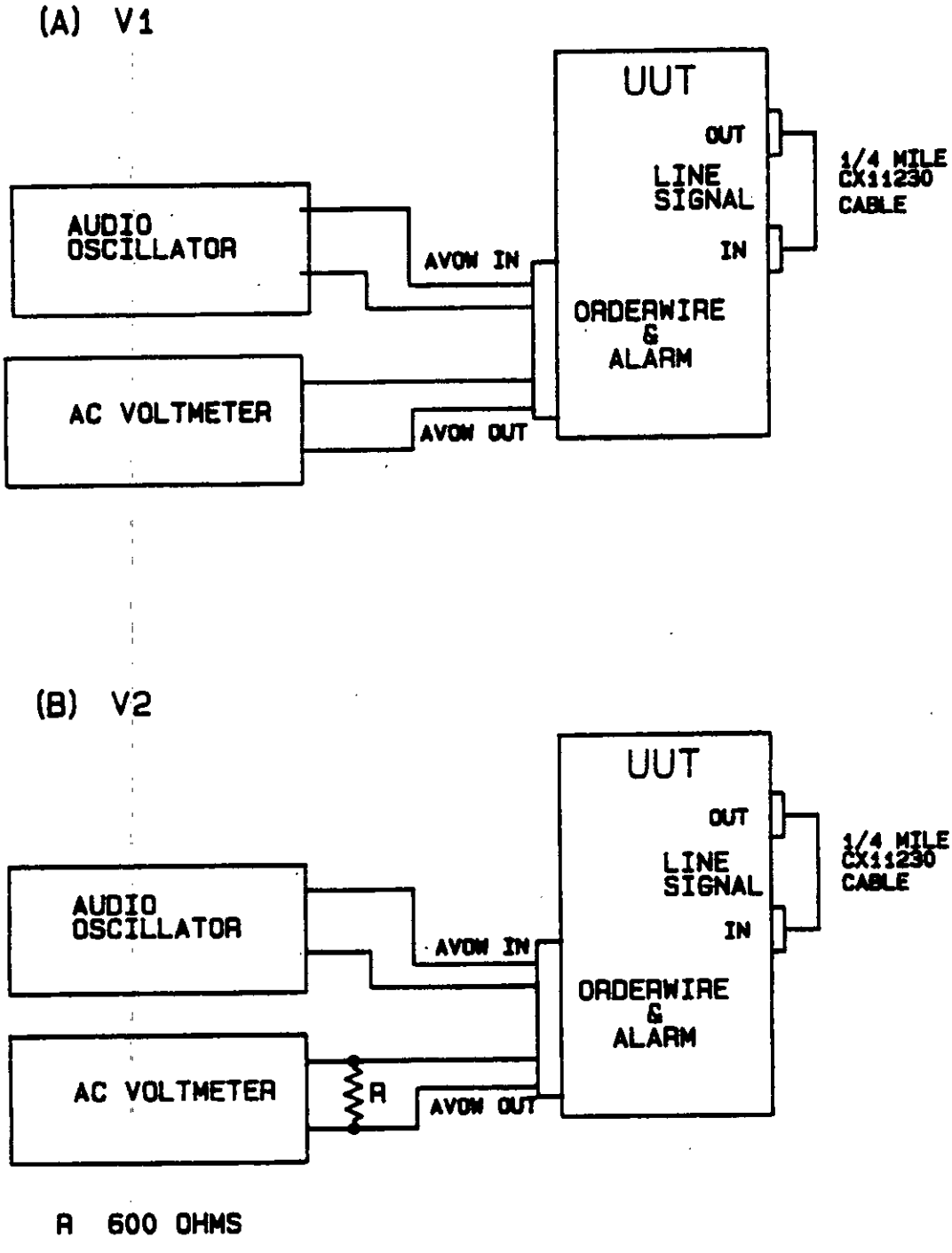
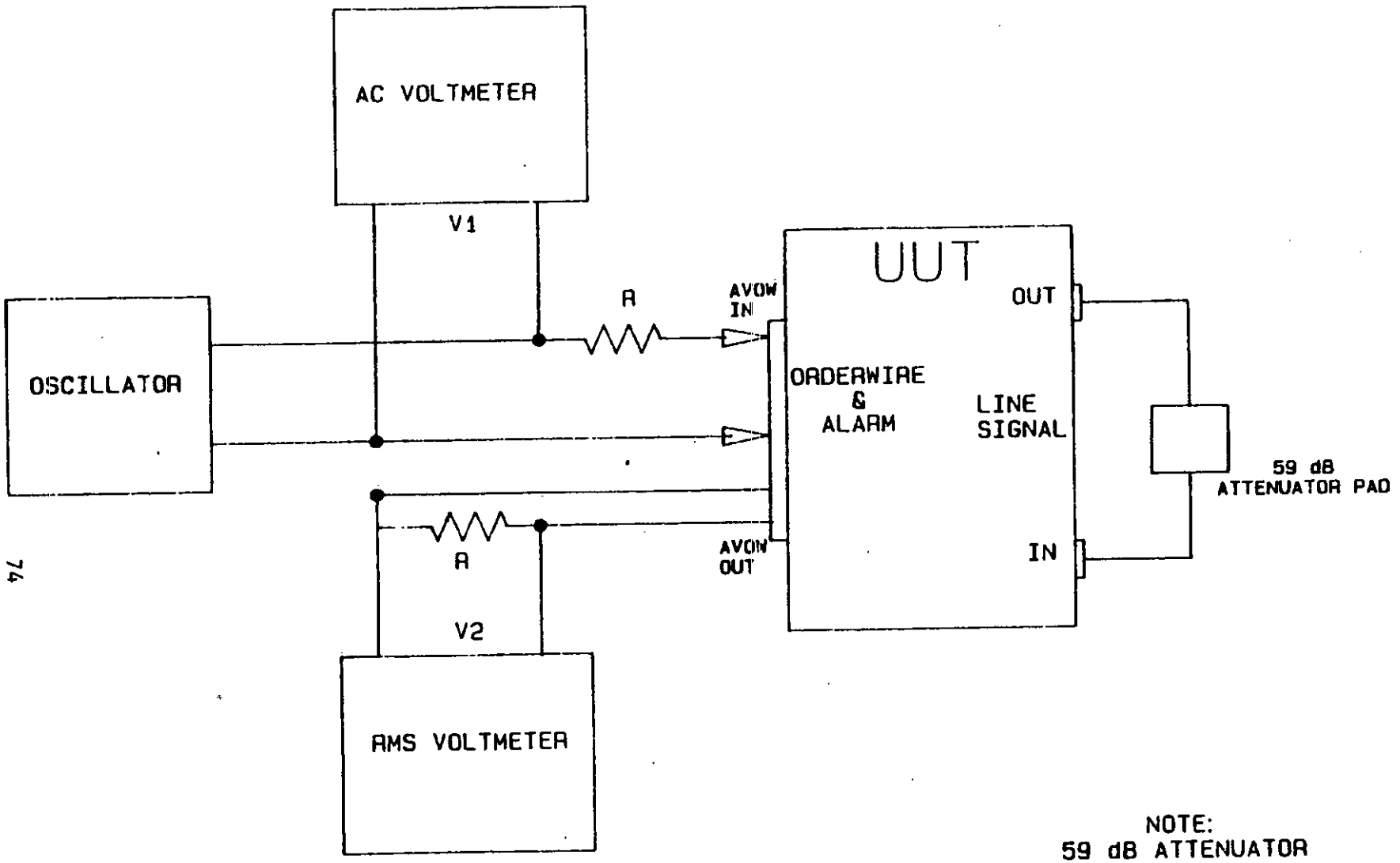


FIGURE 19. Equipment-side maintenance orderwire output impedance.

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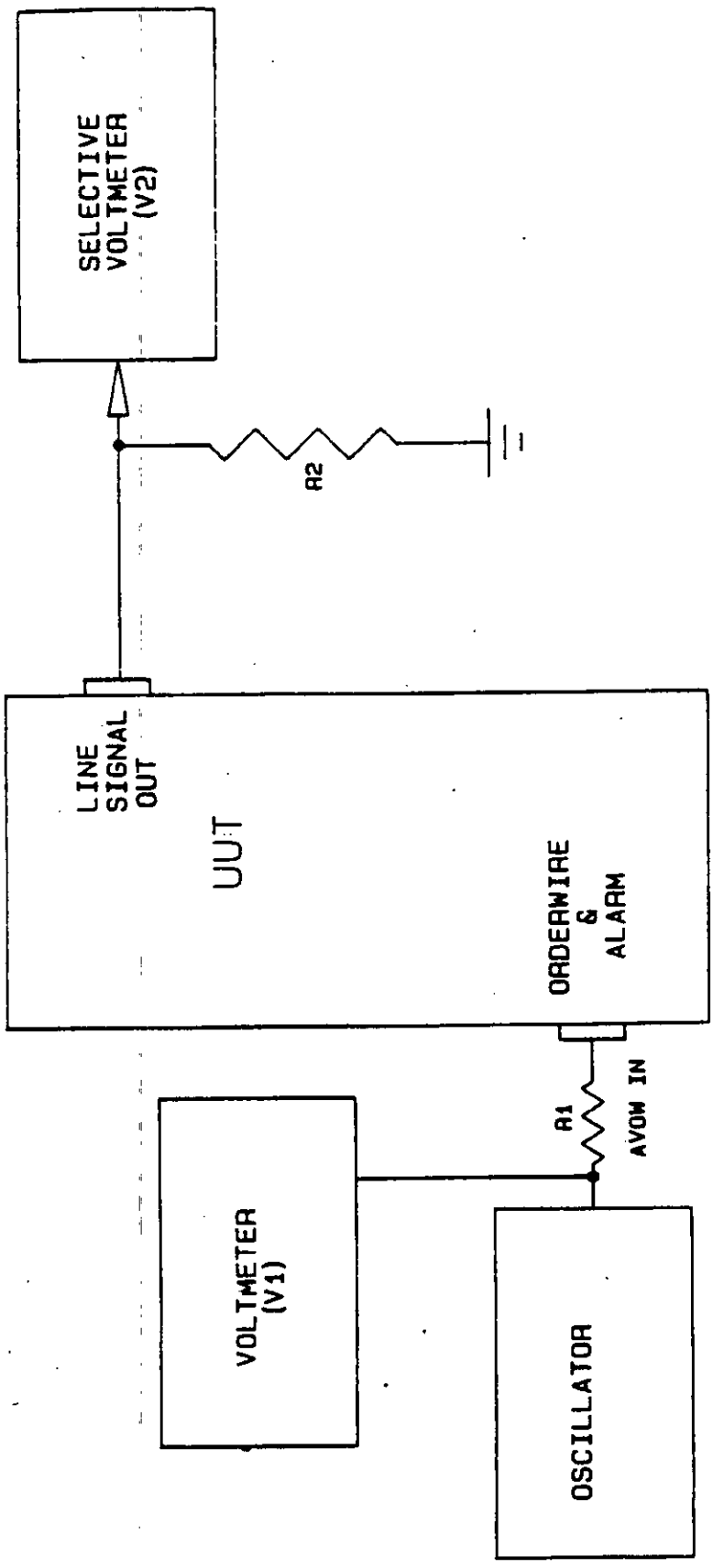
R 600 OHMS ± 1%

CIRCUITRY ADDED TO ORDERWIRE AVC (A6) CCA TO DISABLE AGC.

NOTE:
59 dB ATTENUATOR PAD CAUSES SUMMARY FAULT LAMP AND TRAFFIC RCV A LED TO LIGHT.

FIGURE 20. Frequency response.

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R1 600 OHMS ±1%
R2 430 OHMS ±1%

FIGURE 21. Cable-side maintenance orderwire output power level.

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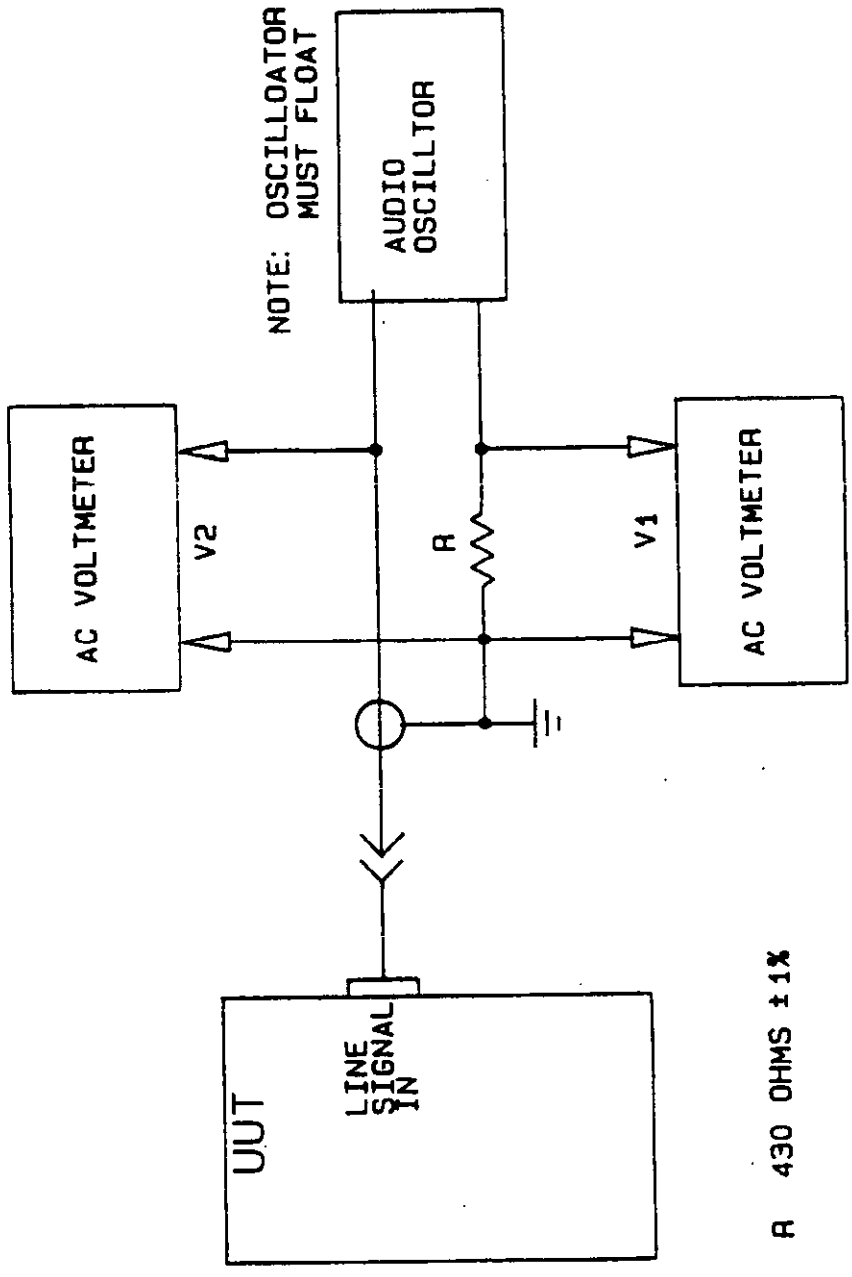
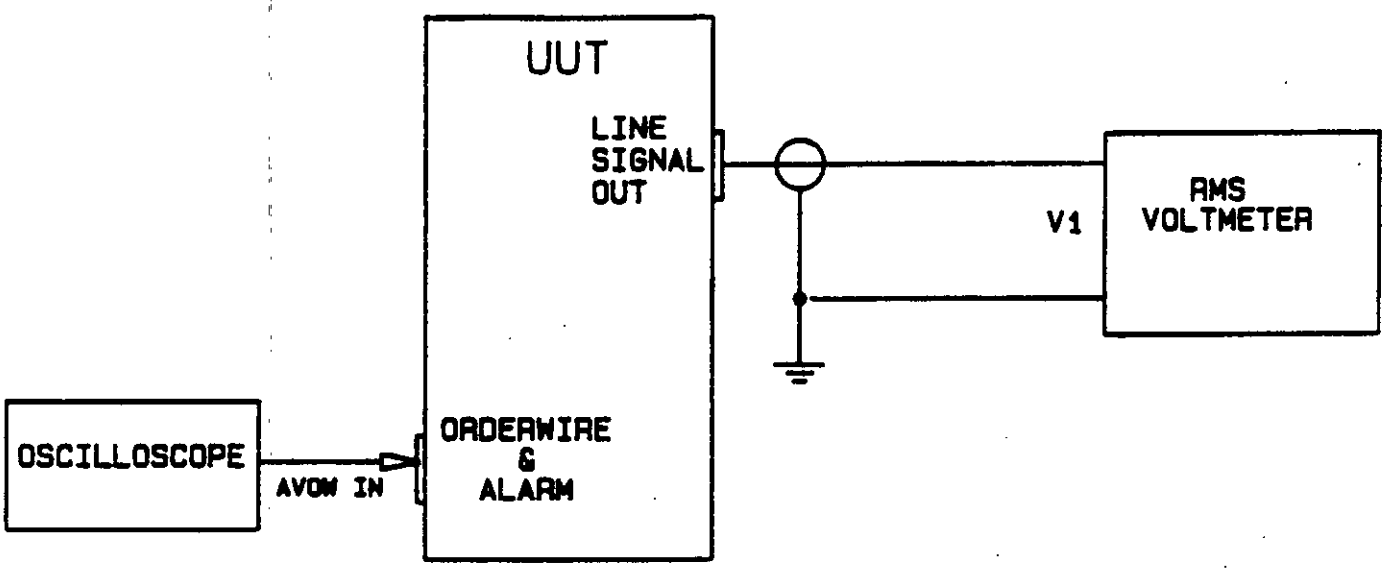


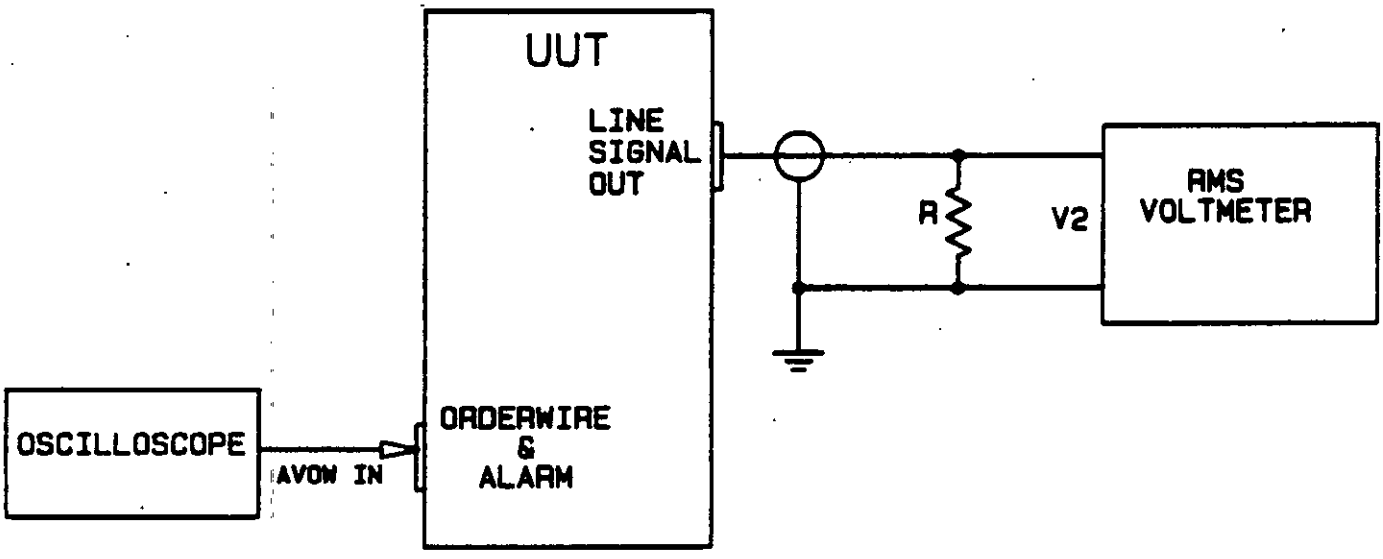
FIGURE 22. Cable-side maintenance orderwire input impedance.

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(A) OPEN CIRCUIT VOLTAGE



(B) LOADED OUTPUT VOLTAGE



R 430 OHMS ± 1%

FIGURE 23. Cable-side maintenance orderwire output impedance.

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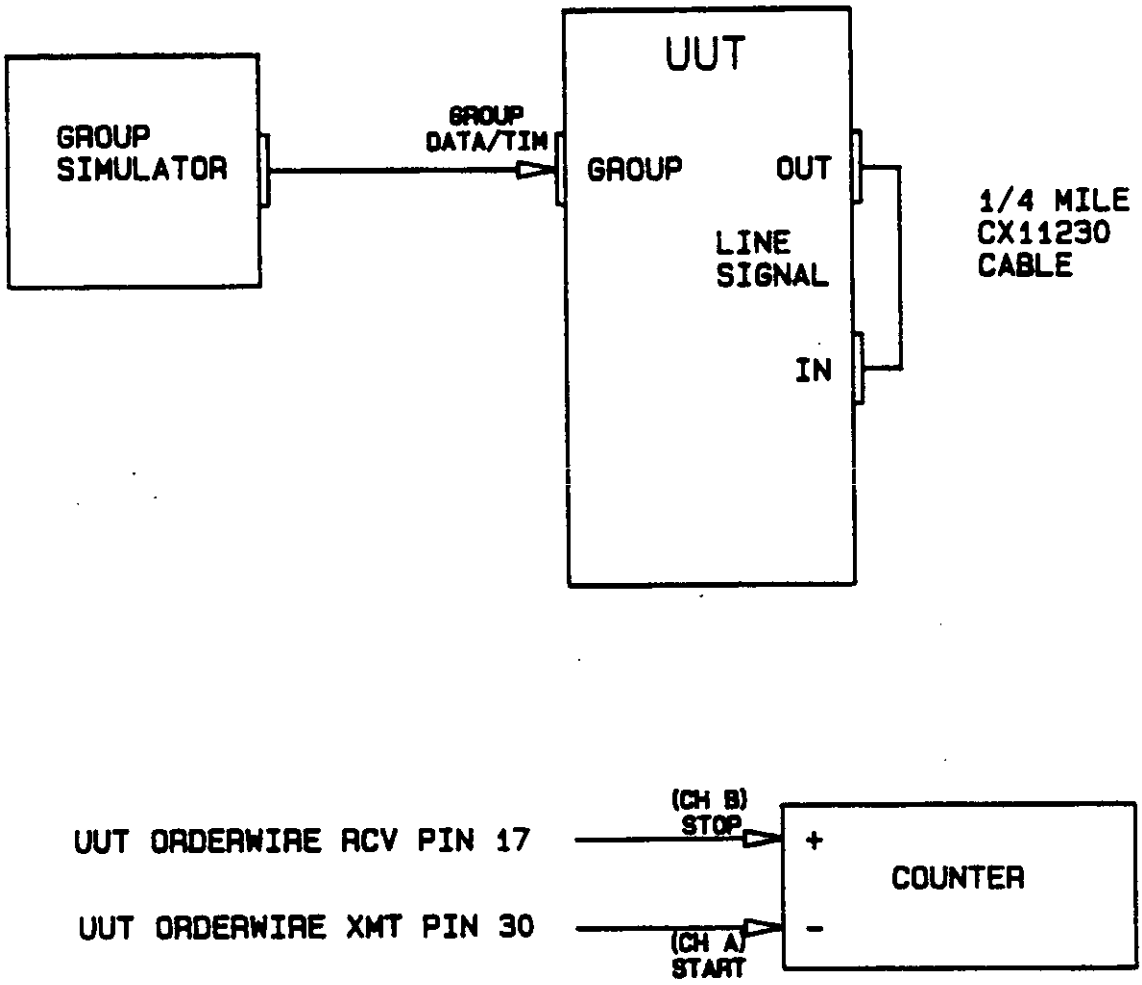


FIGURE 24. Frame acquisition time.

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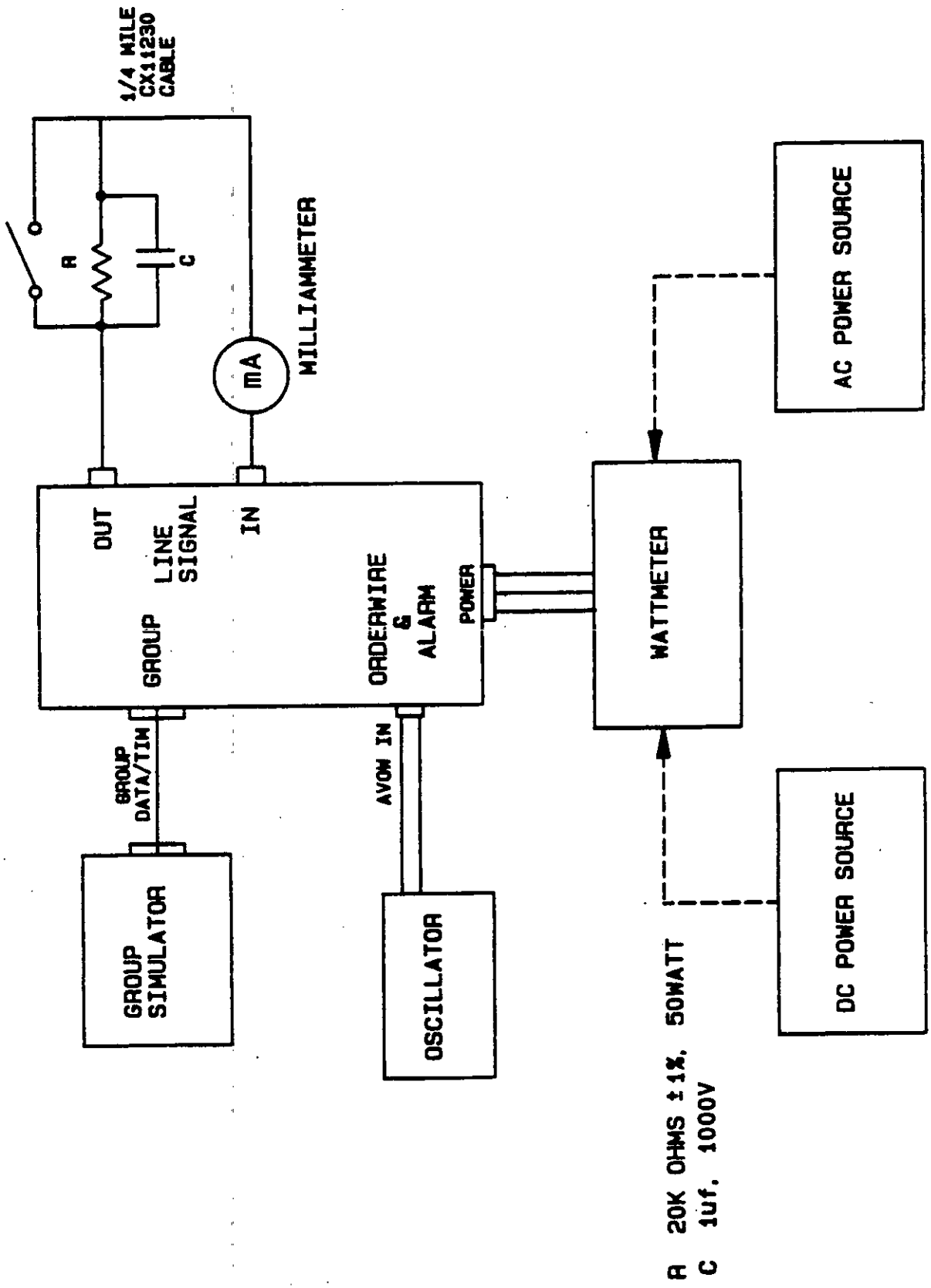
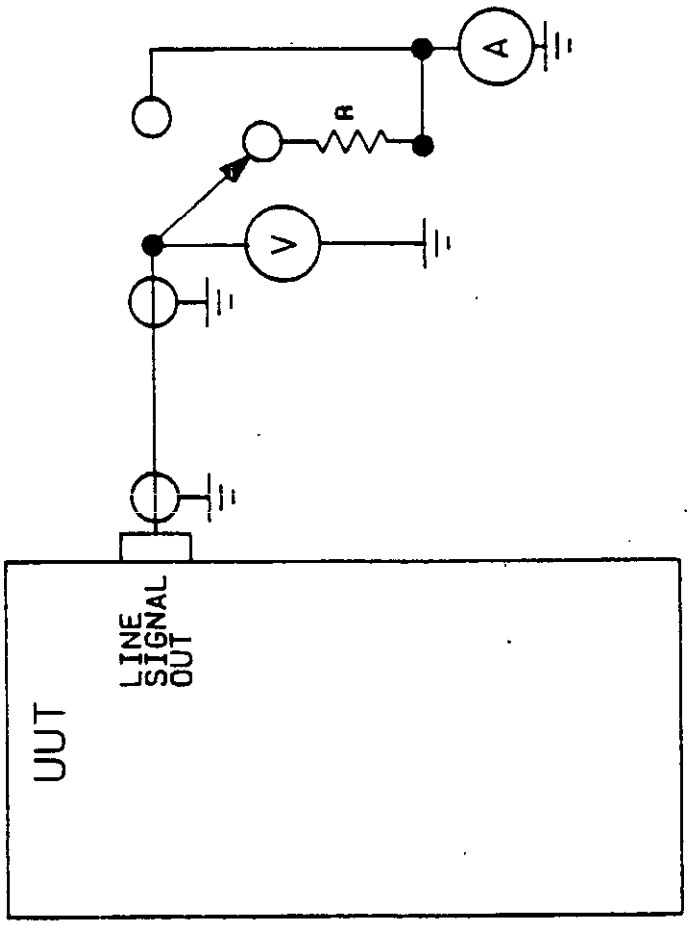


FIGURE 25. Power variation.

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R 20K OHMS ±1%, 50 WATT

FIGURE 26. Output current level.

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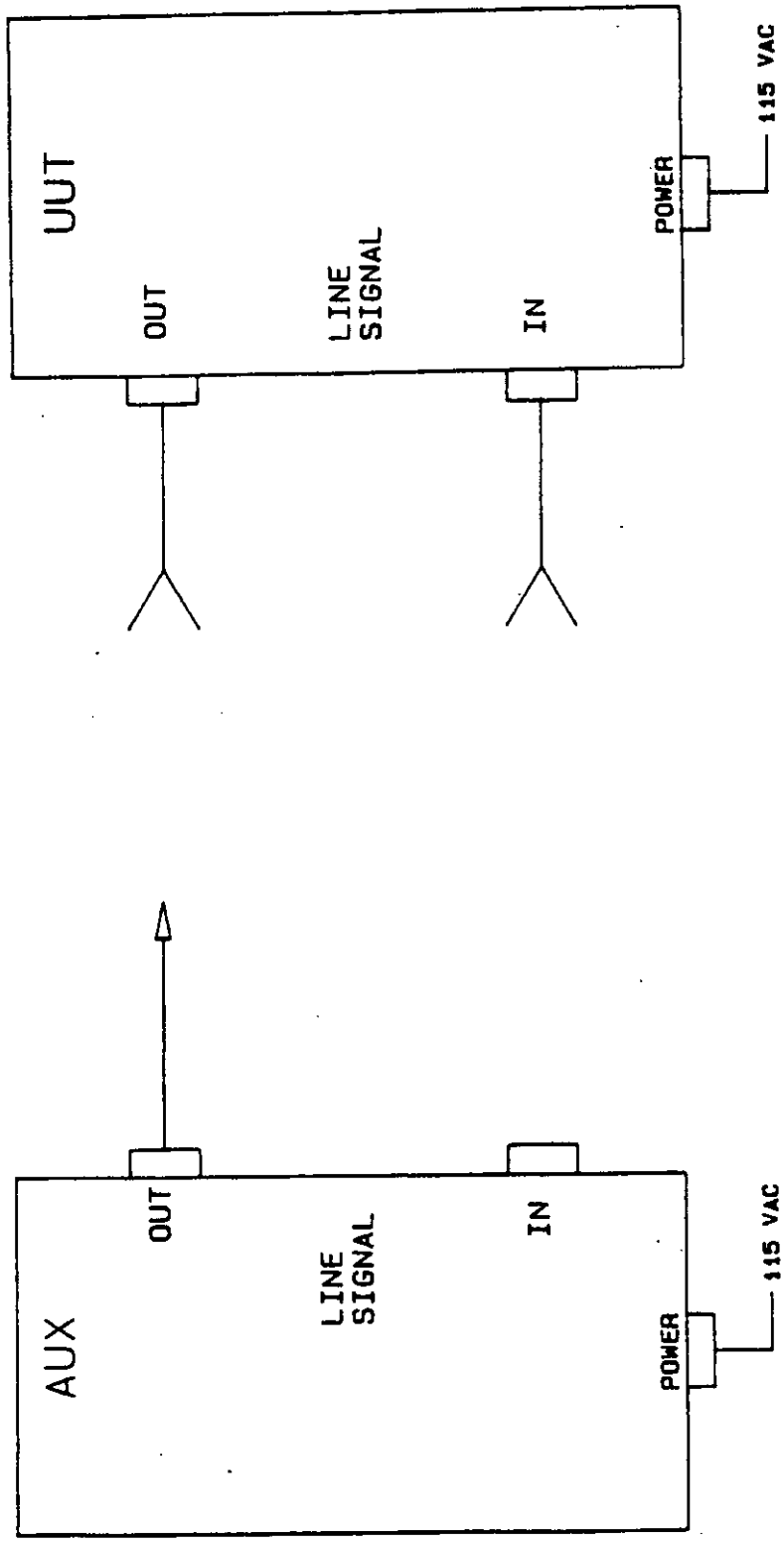


FIGURE 27. Power feed current protection.

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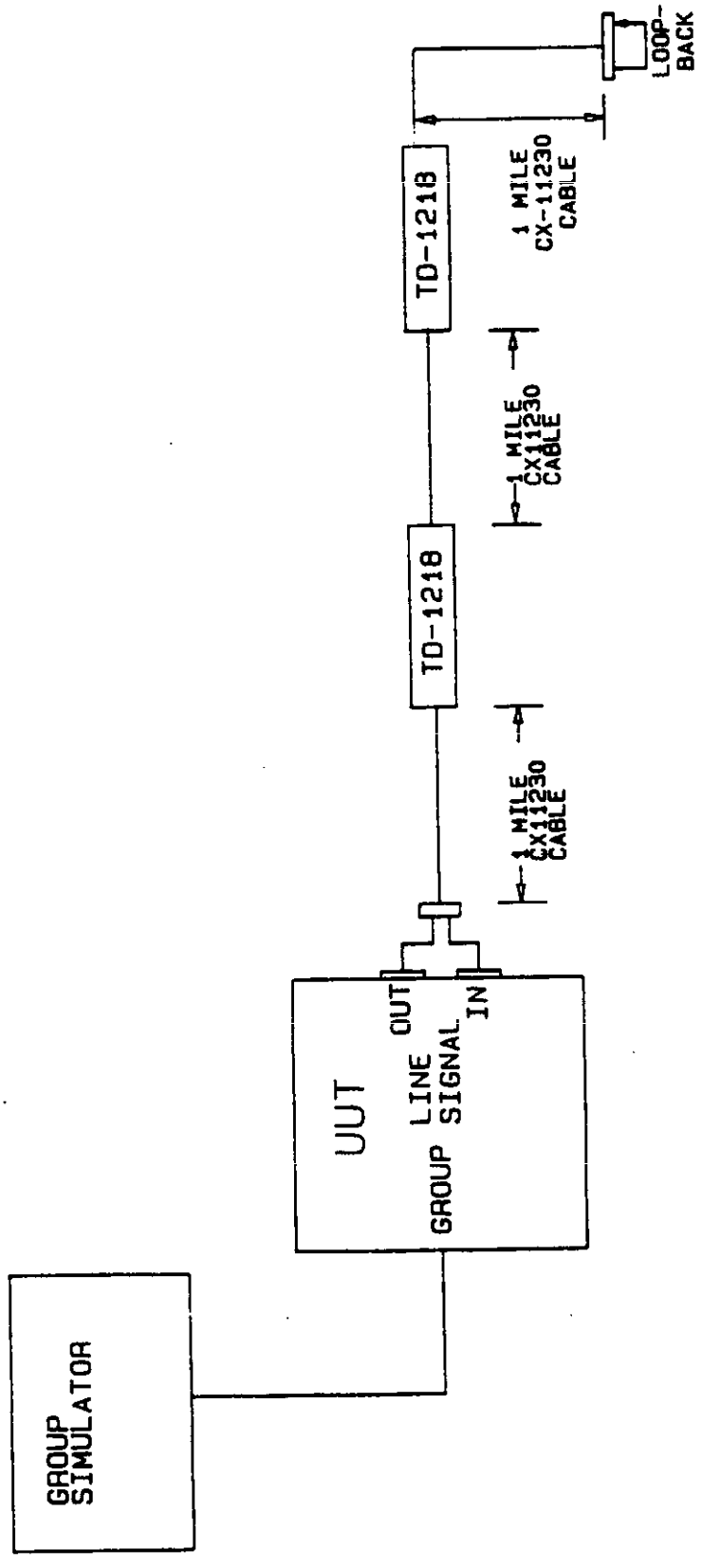


FIGURE 28. Fault locate.

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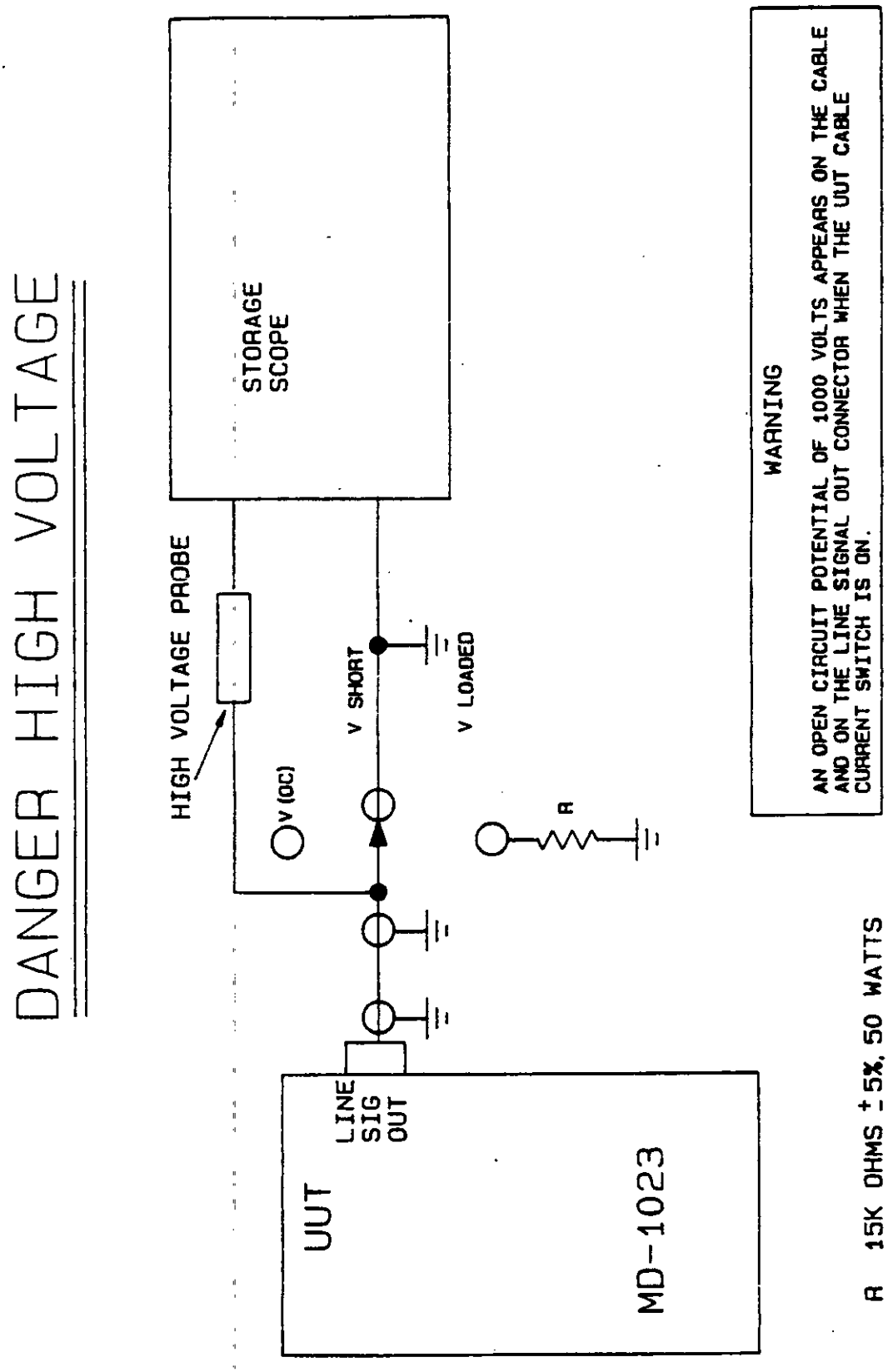
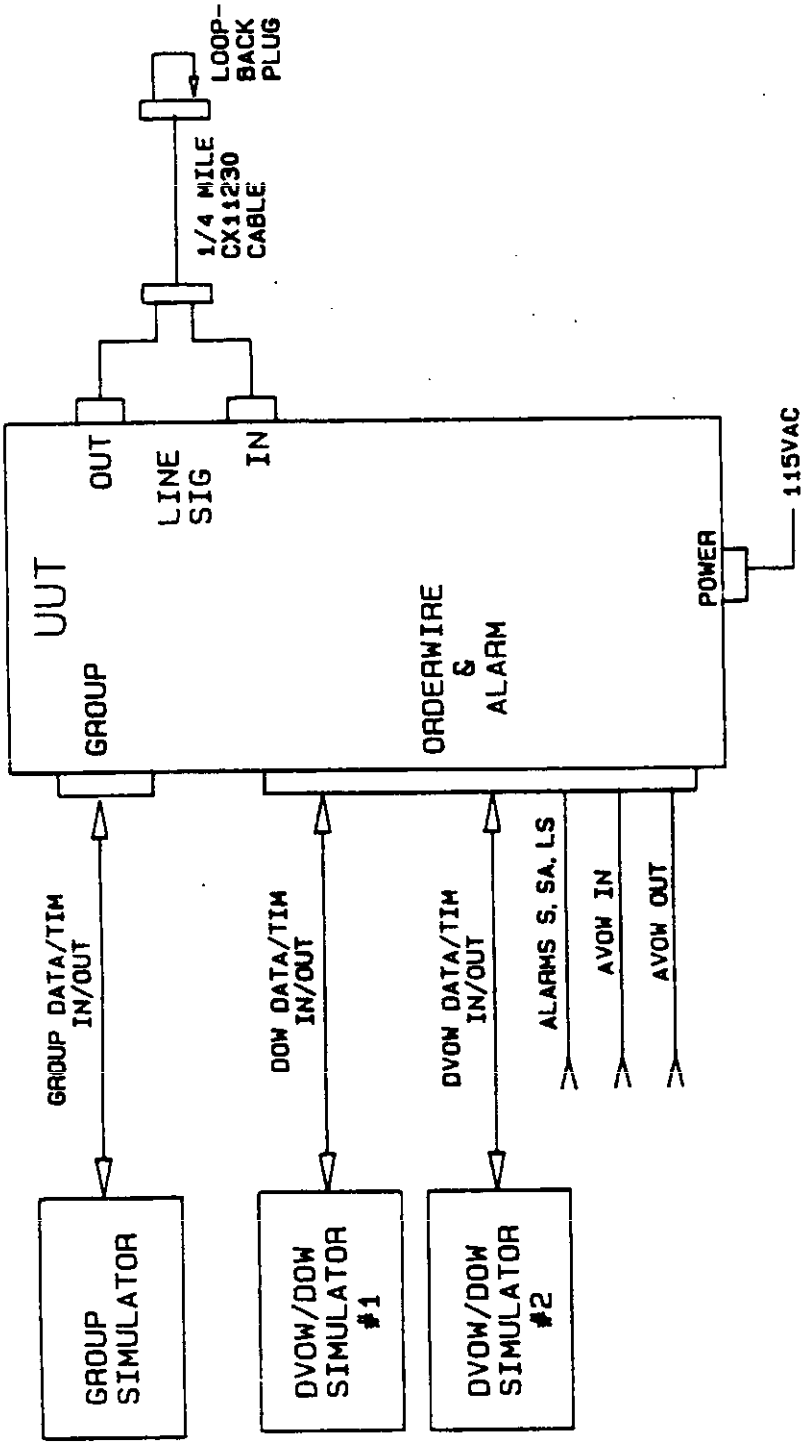
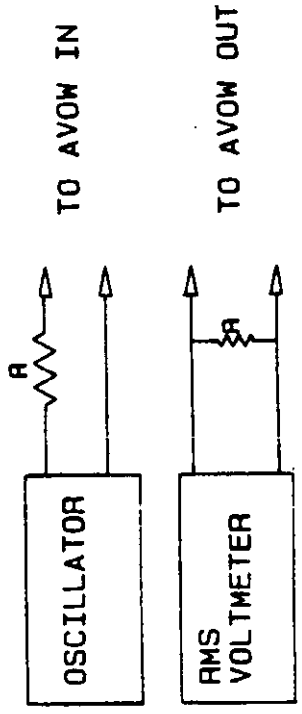
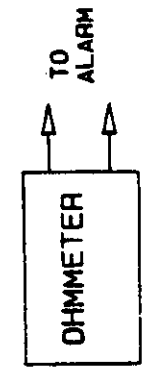


FIGURE 29. Cable-side power feed safty.

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R 600 OHMS ±1%

FIGURE 30. BITE/alarm operational test.

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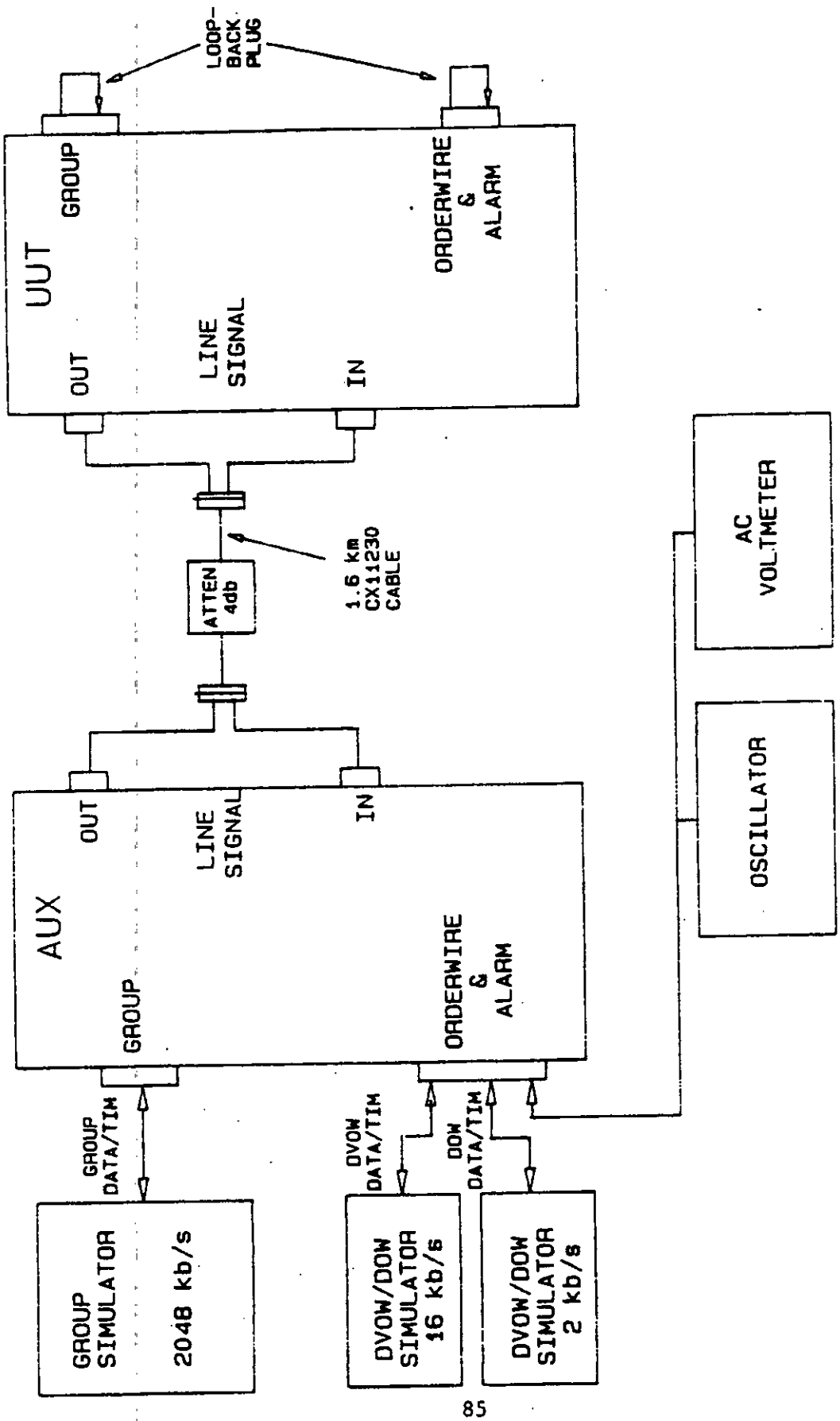


FIGURE 31. Full-duplex signal detection, dynamic range and timing source.

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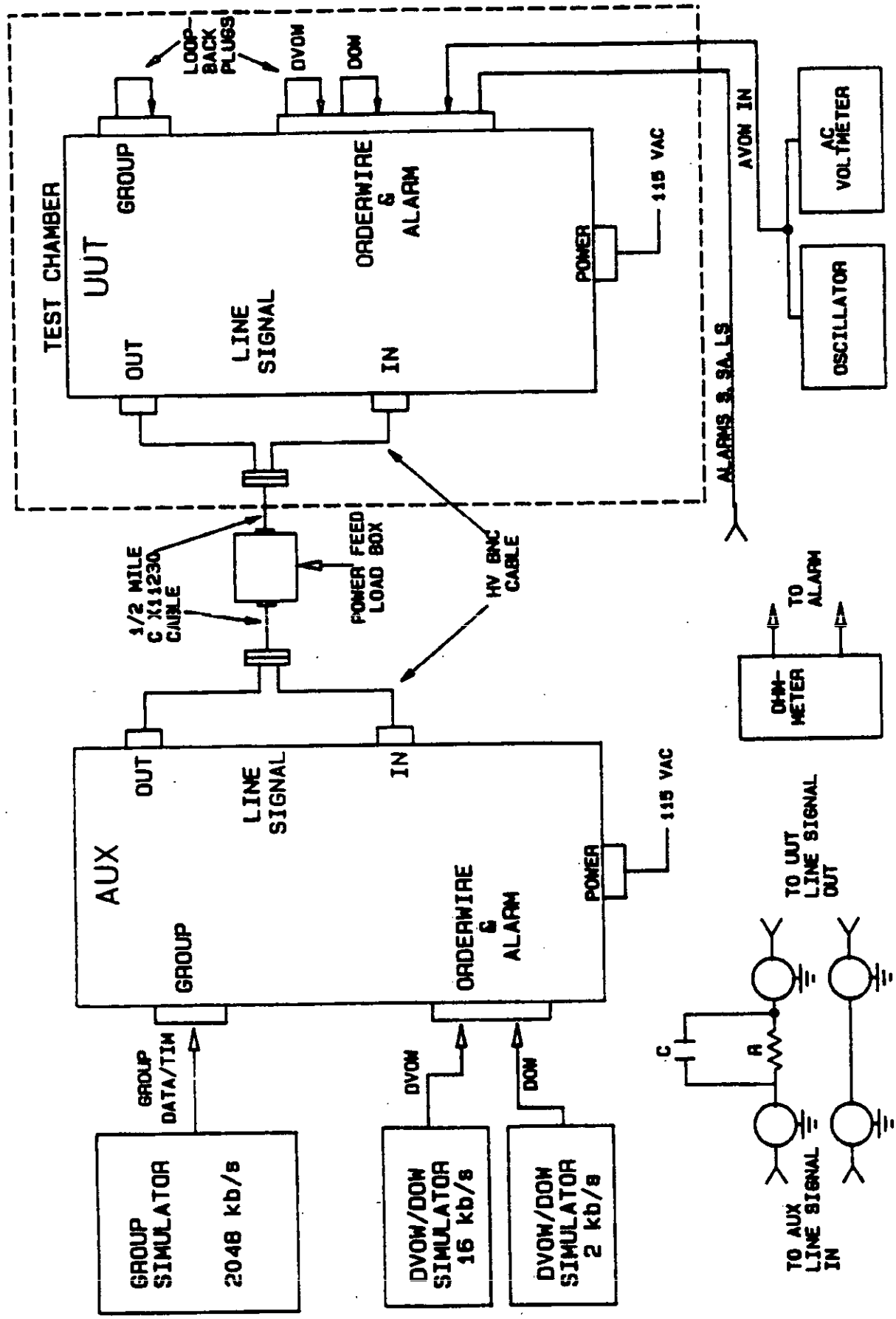


FIGURE 32. Unit performance electrical test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-49231A(CR)	Modem, Cable Driver, MD-1023(P)/G

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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